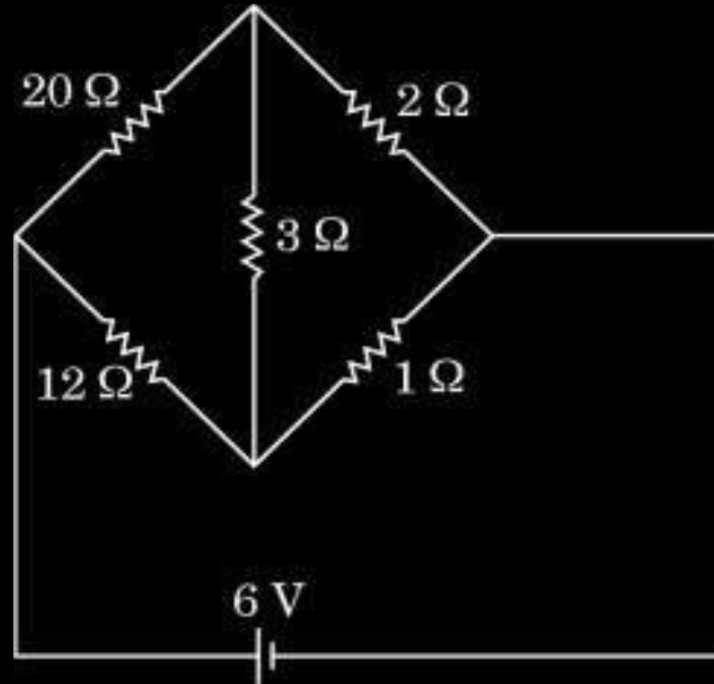
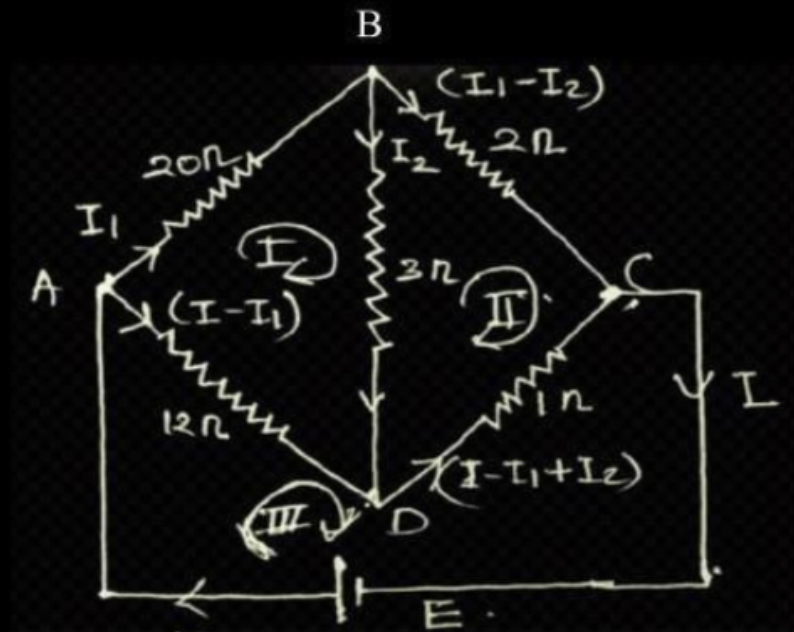


- (ii) Determine the current in $3\ \Omega$ branch of a Wheatstone Bridge in the circuit shown in the figure.





Applying Kirchhoff's rule to loop I (ABDA)

$$-20 I_1 - 3 I_2 + 12 (I - I_1) = 0 \dots\dots\dots(i)$$

Applying Kirchhoff's rule to loop II (BCDB)

$$-2 (I_1 - I_2) + 1 (I - I_1 + I_2) + 3 I_2 = 0 \dots\dots\dots(ii)$$

Applying Kirchhoff's rule loop III (EADCE)

$$-12 (I - I_1) - 1 (I - I_1 + I_2) = -6 \dots\dots\dots(iii)$$

After solving eq. (i), (ii), (iii)...

$$I_2 = 12/821 \text{ A}$$

1/2

1/2

1/2

1/2

1/2

15. *Assertion (A) :* The work done, in taking a unit charge around a closed loop of an electric circuit involving cells and resistors in the loop, is zero.

Reason (R) : The potential at a point depends on the location of the point in the loop. After completing one round, the charge comes back to the point of start.

15. *Assertion (A) :* The work done, in taking a unit charge around a closed loop of an electric circuit involving cells and resistors in the loop, is zero.

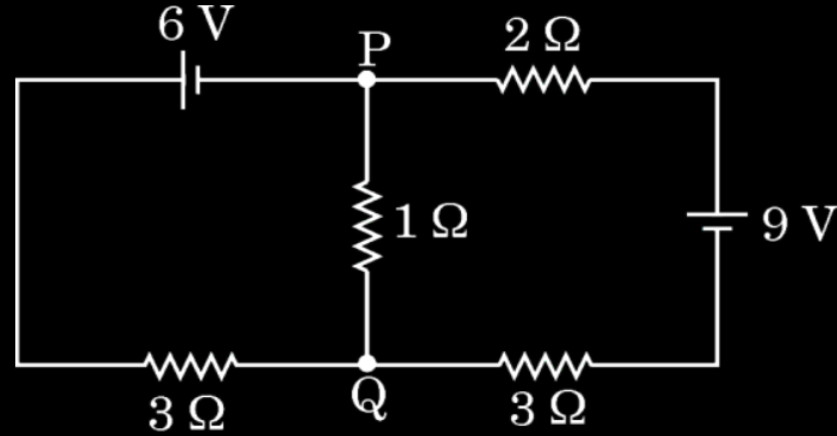
Reason (R) : The potential at a point depends on the location of the point in the loop. After completing one round, the charge comes back to the point of start.

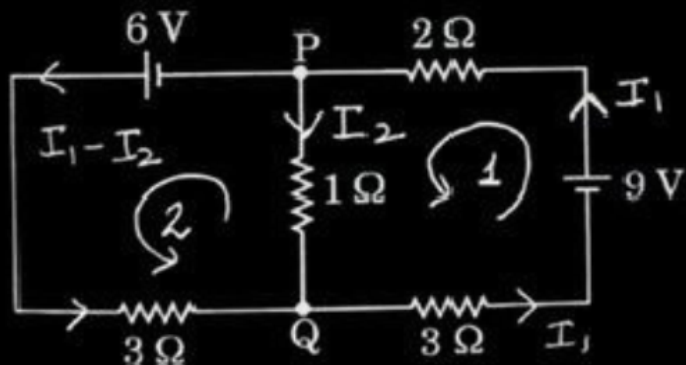
CBSE 26

(A) Both Assertion (A) and Reason (R) are true, and Reason (R) is the correct explanation of Assertion (A).

28. (a) The two Kirchhoff's rules are based on law of conservation of two physical quantities. Name the quantities.
- (b) Find magnitude and direction of current in $1\ \Omega$ resistor in the given circuit.

3





(a)

Junction Rule

It is based on Law of Conservation of Charge.

Loop Rule

It is based on Law of Conservation of Energy.

(b)

Loop equation (1)

$$\begin{aligned} 2I_1 + I_2 + 3I_1 &= 9 \\ 5I_1 + I_2 &= 9 \quad \dots (1) \end{aligned}$$

Loop equation (2)

$$\begin{aligned} 3(I_1 - I_2) - I_2 &= 6 \\ 3I_1 - 3I_2 - I_2 &= 6 \\ 3I_1 - 4I_2 &= 6 \quad \dots (2) \end{aligned}$$

Solving (1) and (2)

$$4(5I_1 + I_2) = 36$$

$$3I_1 - 4I_2 = 6$$

On solving we get,

$$23I_1 = 42 \Rightarrow I_1 = \frac{42}{23} \text{ A}$$

Now, solving for I_2 ,

$$I_2 = 9 - 5I_1 = -\frac{3}{23} \text{ A}$$

Direction of current in 1Ω is from Q to P

$\frac{1}{2}$

$\frac{1}{2}$

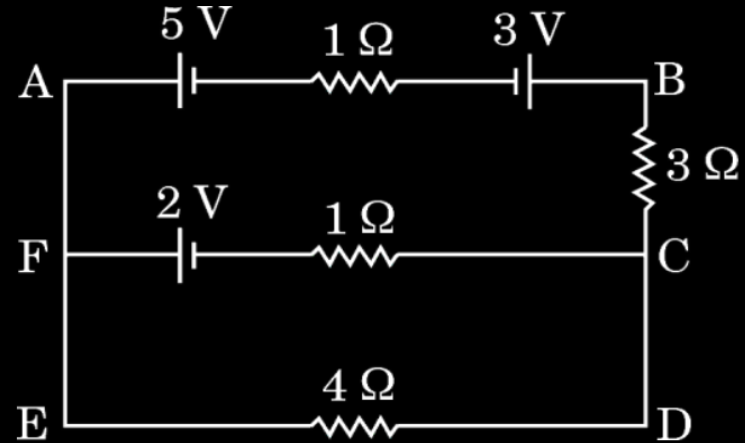
$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

22. State Kirchhoff's rules. Using these rules, find the current flowing through branch FC in the given circuit. 3



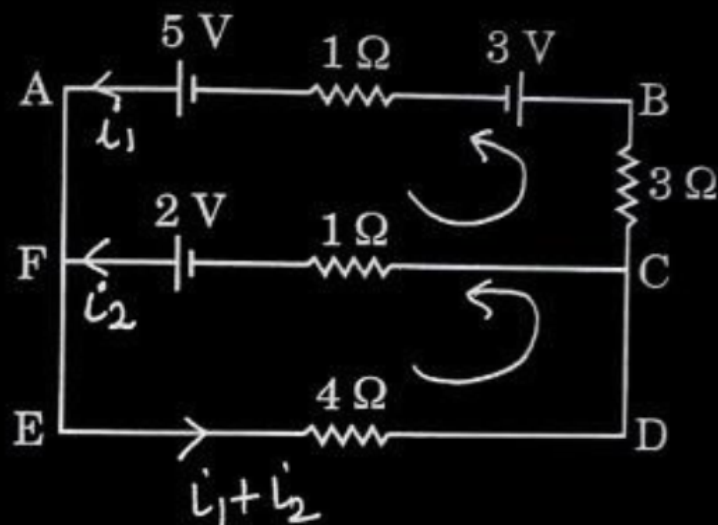
Junction Rule: At any junction, the sum of the currents entering the junction is equal to the sum of the currents leaving the junction.

$\frac{1}{2}$

Loop Rule: The algebraic sum of changes in potential around any closed loop involving resistors and cells in the loop is zero.

$\frac{1}{2}$

Note:- Full credit to be given if stated mathematically.



For loop AFCBA

$$3i_1 + i_1 - i_2 = -3 + 5 - 2$$

$$4i_1 - i_2 = 0$$

$$4i_1 = i_2 \quad \text{-----(i)}$$

For loop FEDCF

$$i_2 + 4(i_1 + i_2) = 2$$

$$4i_1 + 5i_2 = 2 \quad \text{-----(ii)}$$

Solving equations (i) and (ii),

$$i_1 = \frac{1}{12} \text{ A}$$

$$i_2 = \frac{1}{3} \text{ A}$$

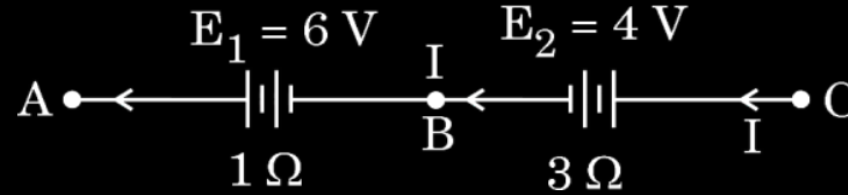
Note: Award full marks if a student solves using another appropriate current distribution in the above circuit.

$\frac{1}{2}$

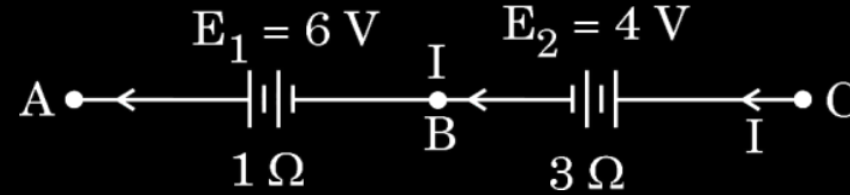
$\frac{1}{2}$

1

20. (a) In the given figure, a steady current I flows through the circuit when points A and C are connected by a wire of negligible resistance. Find the potential difference between points B and C. 2



20. (a) In the given figure, a steady current I flows through the circuit when points A and C are connected by a wire of negligible resistance. Find the potential difference between points B and C. 2



CBSE 26

$$\begin{aligned}
 I &= \frac{\varepsilon_{net}}{R_{net}} \\
 &= \frac{2}{4} \\
 &= 0.5\text{ A}
 \end{aligned}$$

$$\begin{aligned}
 V_{BC} &= -\varepsilon_2 - Ir_2 \\
 &= -4 - (0.5 \times 3) \\
 &= -5.5\text{ V}
 \end{aligned}$$

$\frac{1}{2}$

$\frac{1}{2}$

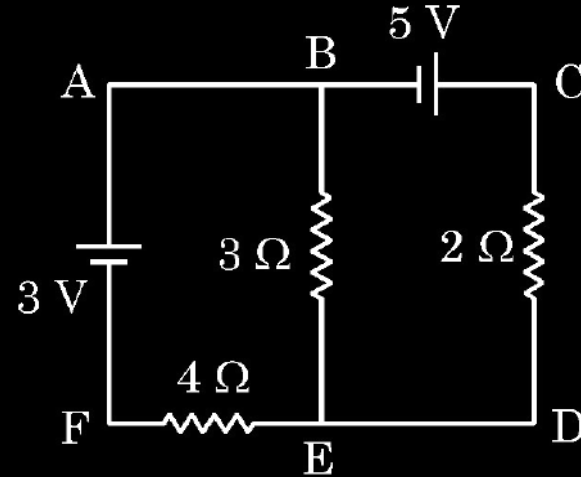
$\frac{1}{2}$

$\frac{1}{2}$

27. (a) Explain the statement : “Current is a scalar although we represent current with an arrow”.

3

(b) Use Kirchhoff's rules to find the current through $3\ \Omega$ resistor in the circuit shown in the figure :



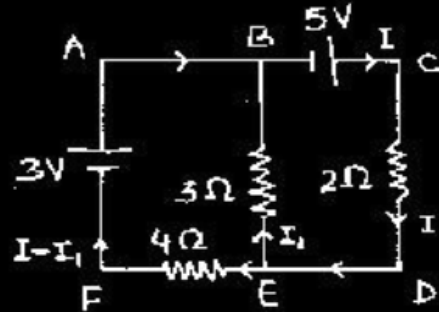
- (a) Electric current is a scalar quantity because it does not obey law of vector addition.
The current with an arrow is only to signify the direction of conventional current.

1

Alternatively:

Current do not follow the law of vector addition. The current through an area of cross-section is given by the scalar product of two vectors. $I = \vec{j} \cdot \vec{\Delta S}$

(b)



$1/2$

In Loop ABEFA

$$3 + 3I_1 - 4(I - I_1) = 0$$

$$4I - 7I_1 = 3 \dots \dots \dots (1)$$

$1/2$

In loop BCDEB

$$5 - 2I - 3I_1 = 0$$

$$2I + 3I_1 = 5 \dots \dots \dots (2)$$

$1/2$

Solving equations (1) and (2), we get

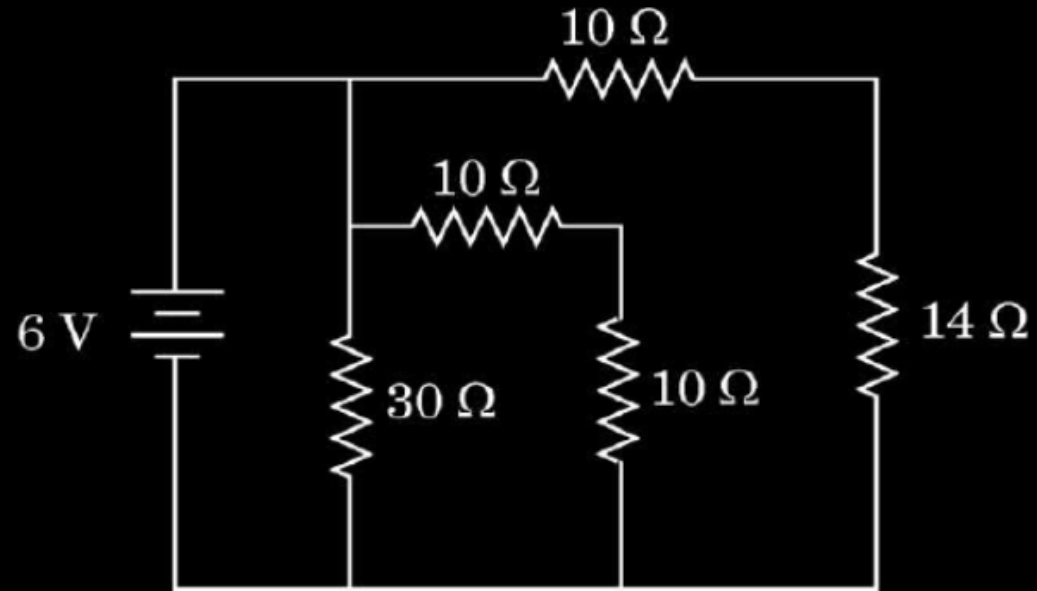
Current through 3Ω resistor is

$$I_1 = \frac{7}{13} A$$

$1/2$

Note: Award full marks if a student solves by another appropriate current distribution in the circuit.

- (iii) Consider the circuit shown in the figure. Find the effective resistance of the circuit and the current drawn from the battery.



(iii) From the diagram

10Ω and 14 Ω are in series $R_1=10\ \Omega + 14\ \Omega = 24\ \Omega$

10 Ω and 10 Ω are in series $R_2=10\ \Omega + 10\ \Omega = 20\ \Omega$

24 Ω, 20 Ω and 30 Ω are in parallel.

$$\frac{1}{R} = \frac{1}{24} + \frac{1}{20} + \frac{1}{30} = \frac{5 + 6 + 4}{120} = \frac{15}{120}$$

$$R = 8\ \Omega$$

Electric current in the circuit

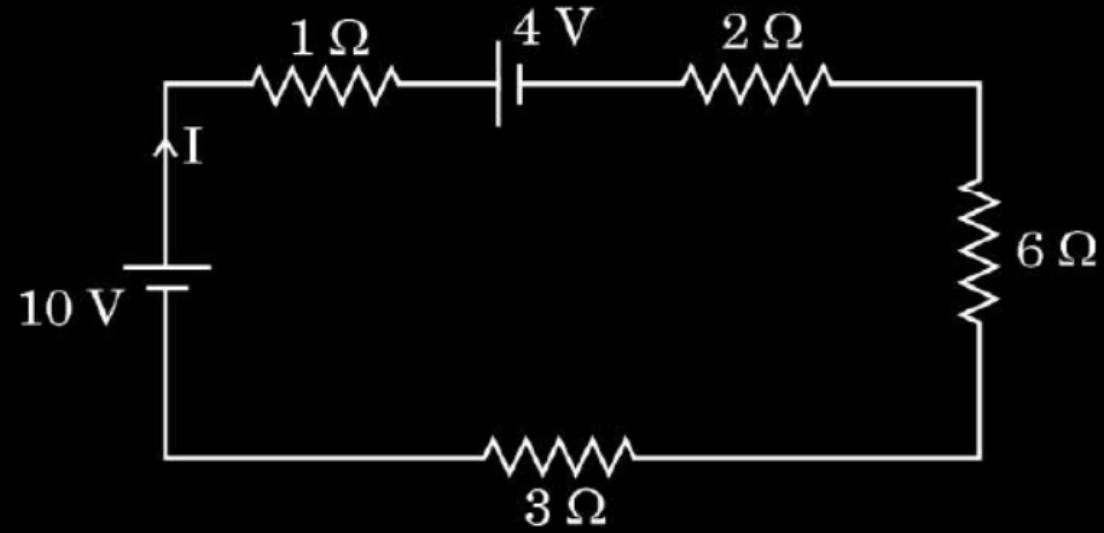
$$I = \frac{V}{R} = \frac{6}{8} = \frac{3}{4} A$$

Note: Full credit of 1 ½ marks is to be awarded for part (iii) even if a student does not attempt this part.

½

1

- (iii) Calculate the total energy supplied by the batteries to the circuit shown in the figure, in one minute.



(iii) Applying Kirchhoff's loop rule

$$10 - I \times 1 - 4 - 2I - 6I - 3I = 0$$

$$12I = 6 \Rightarrow I = 0.5 \text{ A}$$

$$\text{Heat energy } H = I^2 R t$$

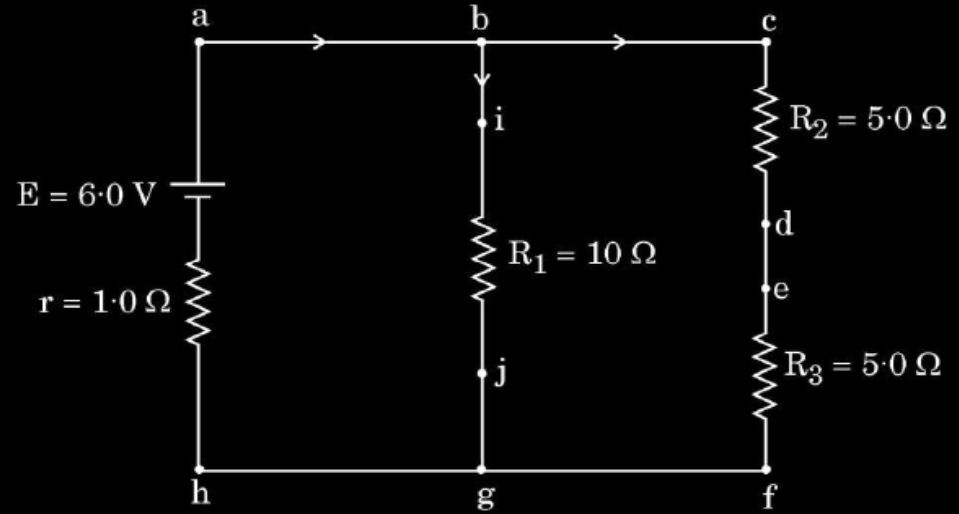
$$H = 0.25 \times 12 \times 60 = 180 \text{ J}$$

$$\frac{1}{2}$$

$$\frac{1}{2}$$

$$\frac{1}{2}$$

34. The following figure shows a circuit diagram. We can find the currents through and potential differences across different resistors using Kirchhoff's rules.



Answer the following questions based on the above :

- | | |
|---|---|
| (a) Which points are at the same potential in the circuit ? | 1 |
| (b) What is the current through arm bg ? | 1 |
| (c) Find the potential difference across resistance R_3 . | 2 |

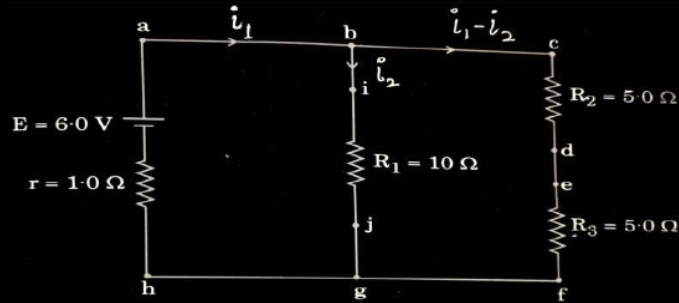
OR

- | | |
|--|---|
| (c) What is the power dissipated in resistance R_2 ? | 2 |
|--|---|

a) Points (a, b, c)
(d, e)
(j, f, g, h)
are at same potential

Note: Give full credit if a student mentions any two points at same potential from the above.

b)



According to Kirchhoff's loop rule

for closed loop abgha

$$-6 + 10 I_2 + I_1 = 0$$

$$I_1 + 10 I_2 = 6 \quad (i)$$

for closed loop acfha

$$-6 + 10 (I_1 - I_2) + I_1 = 0$$

$$11 I_1 - 10 I_2 = 6 \quad (ii)$$

Adding (i) and (ii)

$$12 I_1 = 12$$

$$I_1 = 1 \text{ A}$$

$$I_2 = 0.5 \text{ A}$$

= current through arm bg

Note: Award 1 mark if a student calculates the current by any other method.

1

$$\begin{aligned} c) V_{R3} &= (I_1 - I_2) \times R_3 \\ &= 0.5 \times 5 \\ &= 2.5 \text{ V} \end{aligned}$$

1

1

OR

$$\begin{aligned} (c) P &= (I_1 - I_2)^2 \times R_2 = (0.5)^2 \times 5 \\ &= 1.25 \text{ W} \end{aligned}$$

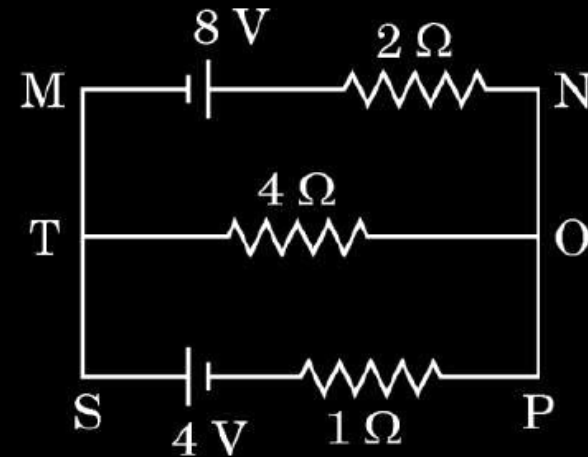
1

1

1/2

1/2

- (ii) Use Kirchhoff's rule to determine the currents flowing through the branches MN, TO and SP in the circuit shown in the figure.



(i) Kirchhoff's junction rule - at any junction, the sum of the current entering the junction is equal to the sum of currents leaving the junction.

Kirchhoff second rule:

The algebraic sum of changes in potential around any closed loop involving resistors and cells in the loop is zero.

In balanced bridge $I_g=0$,

Hence $I_1 = I_3$ and $I_2 = I_4$

Using Kirchhoff's loop rule for closed loops ADBA and CBDC

$$-I_1 R_1 + 0 + I_1 R_1 = 0 \quad (I_g=0) \quad \text{-----(1)}$$

In the second loop $I_3 = I_1$, $I_4 = I_2$

$$I_2 R_4 + 0 - I_1 R_3 = 0 \quad \text{-----(2)}$$

From equation (1) and (2)

$$\frac{I_1}{I_2} = \frac{R_2}{R_1} \text{ and } \frac{I_1}{I_2} = \frac{R_4}{R_3}$$

$$\frac{R_2}{R_1} = \frac{R_4}{R_3}$$

This is the condition for balanced Wheatstone bridge

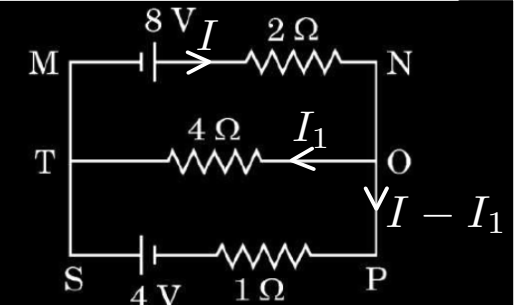
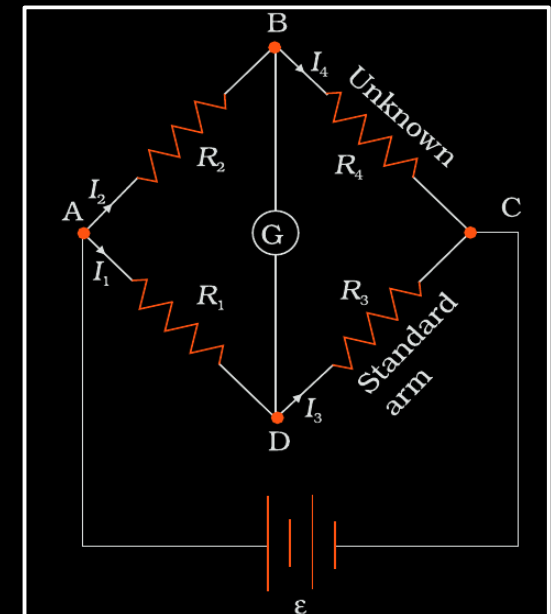
$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$



(ii) In loop MNOTM

$$2I + 4I_1 = 8 \quad \text{-----(1)}$$

Loop OPSTO

$$-I + 5I_1 = -4 \quad \text{-----(2)}$$

On solving

Current in MN, $I = 4A$

Current in TO, $I_1 = 0A$

Current in SP, $I - I_1 = 4A$

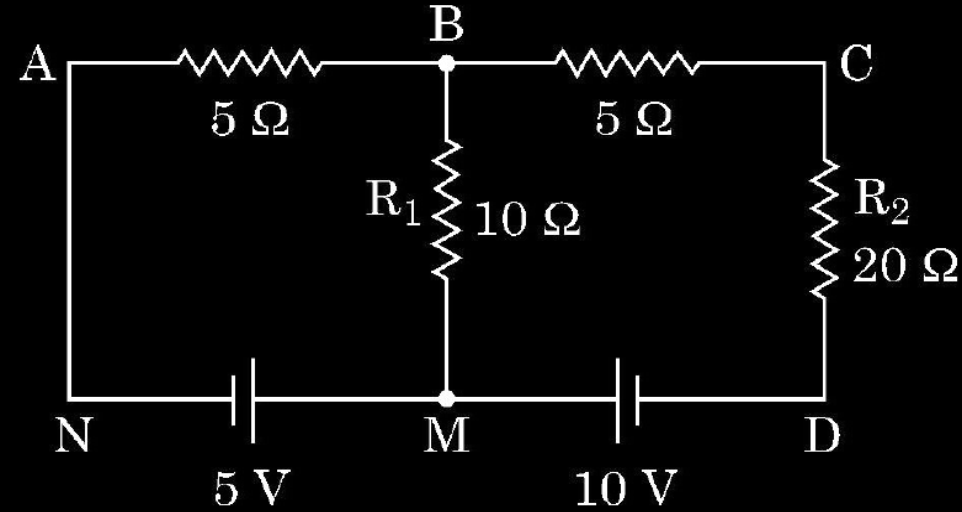
$\frac{1}{2}$

$\frac{1}{2}$

1

27. Find the currents flowing through the branches AB and BC in the network shown.

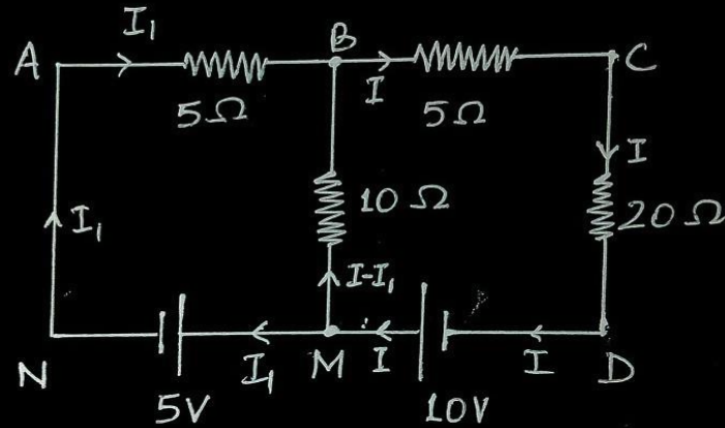
3



Finding current in the arm AB
Finding current in the arm BC

$1\frac{1}{2}$
 $1\frac{1}{2}$

Circuit diagram with distribution of current



1

Using Kirchhoff's voltage rule

In closed loop ABMNA,

$$-5I_1 + 10(I - I_1) - 5 = 0 \quad \dots\dots\dots (1)$$

$\frac{1}{2}$

In closed loop ACDNA

$$-5I - 20I + 10 - 5 - 5I_1 = 0 \quad \dots\dots\dots (2)$$

$\frac{1}{2}$

Solving eq (1) and (2)

$$I_1 = -\frac{3}{17} \text{ A and } I = \frac{4}{17}$$

$$\text{Magnitude of current in arm AB} = \frac{3}{17} \text{ A}$$

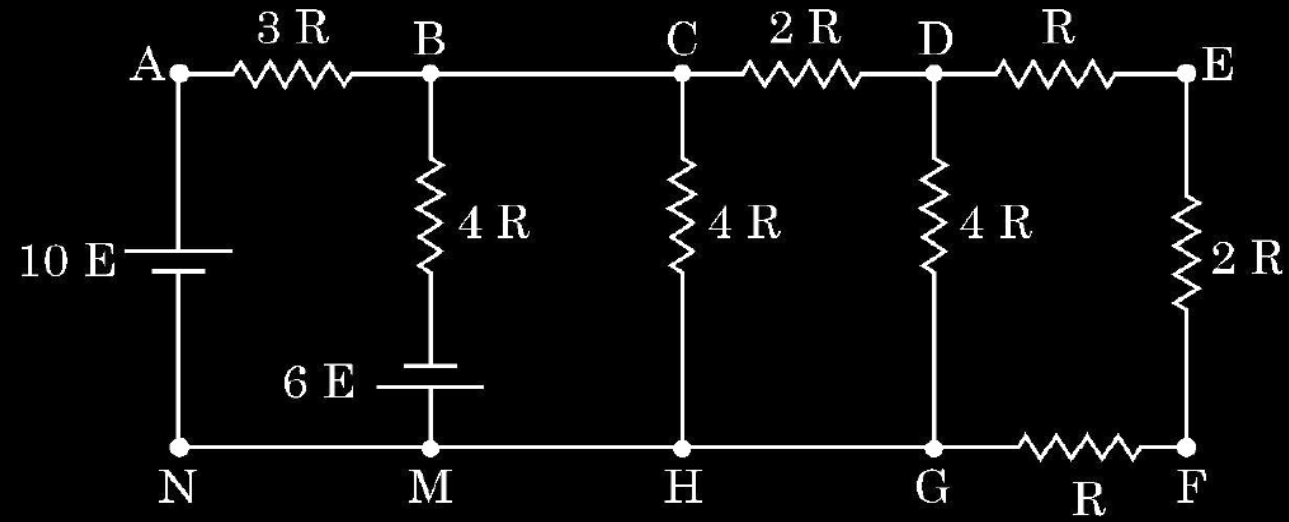
$\frac{1}{2}$

$$\text{Magnitude of current in arm BC} = \frac{4}{17} \text{ A}$$

$\frac{1}{2}$

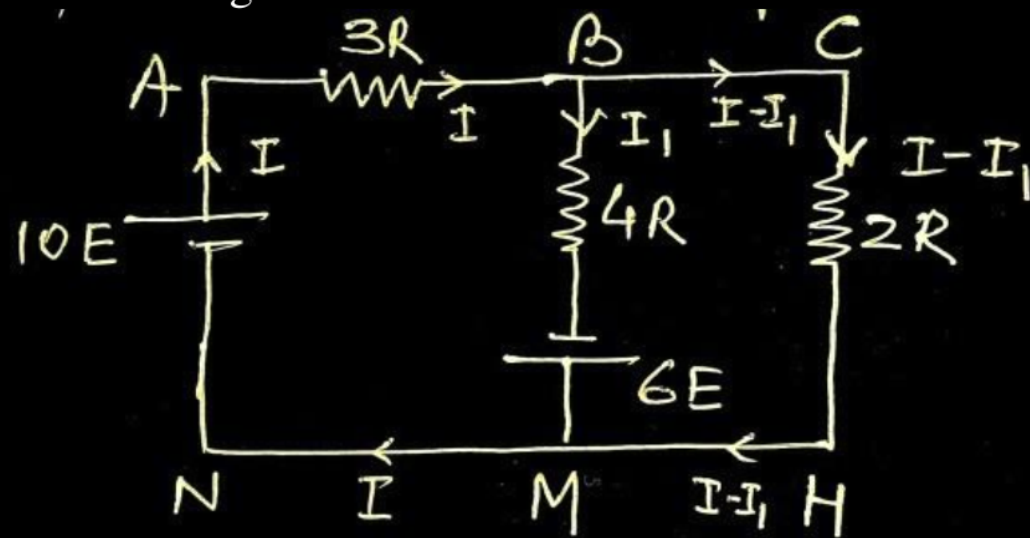
25. Find the current in branch BM in the network shown :

3



Finding equivalent resistance across CH, $R_{CH} = 2R$

Equivalent circuit diagram



In closed loop ABMNA

$$-3IR - 4I_1R + 16E = 0 \quad \dots\dots\dots(1)$$

In closed loop BCHMB

$$-2R(I - I_1) - 6E + 4I_1R = 0 \quad \dots\dots\dots(2)$$

On solving equations (1) and (2)

$$I_1 = \frac{25E}{13R}$$

$\frac{1}{2}$

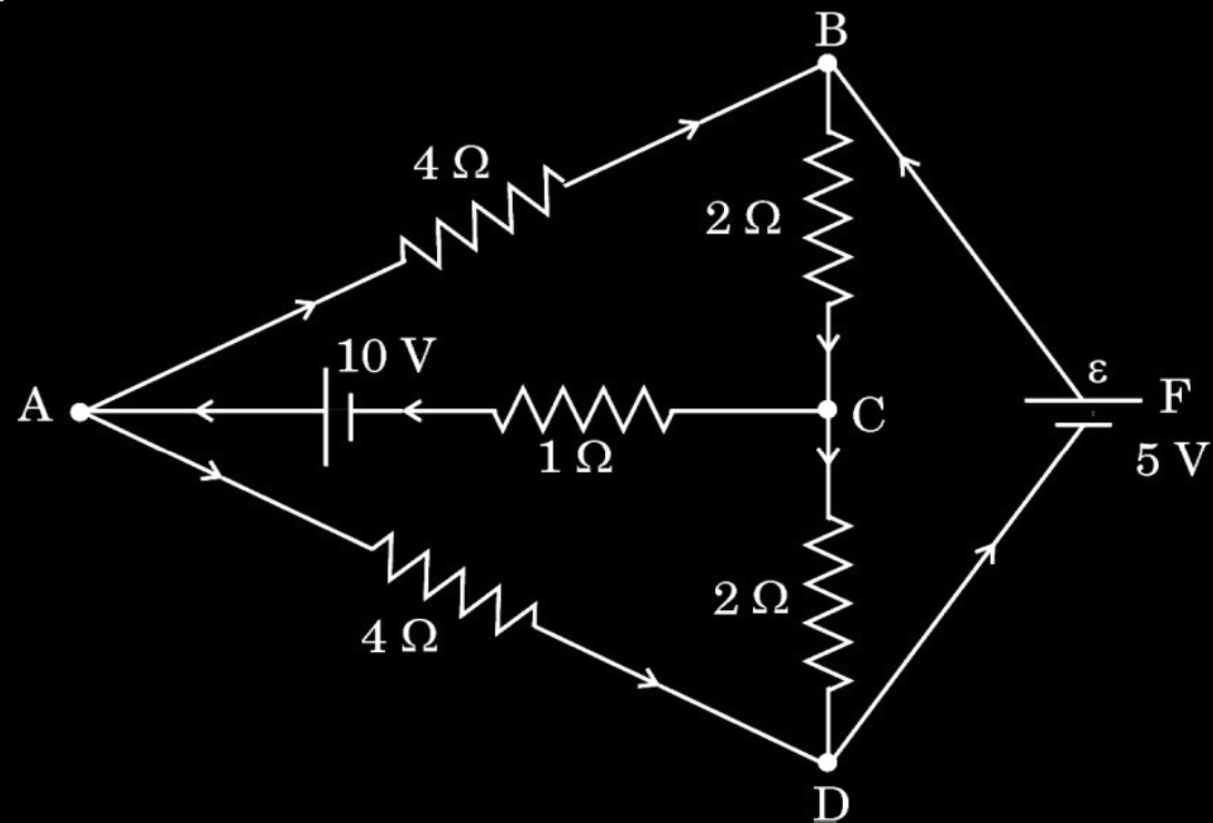
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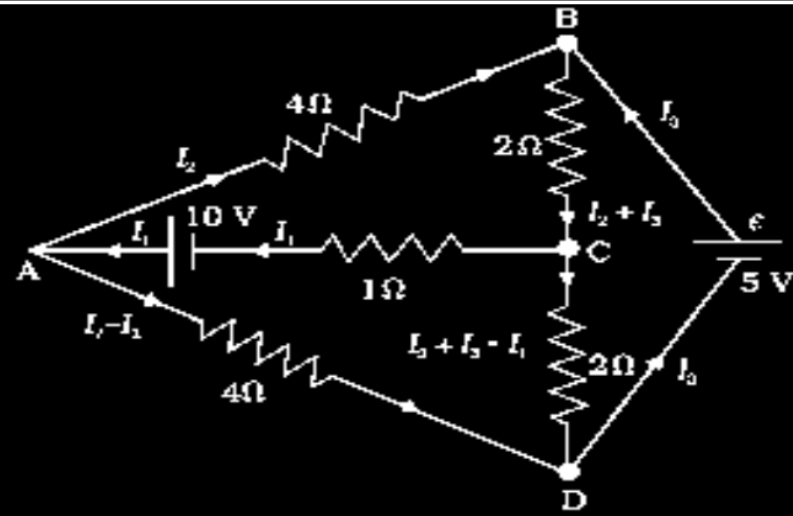
$\frac{1}{2}$

$\frac{1}{2}$

1

22. Determine the current in branches AB, AC and BC of the network shown in figure.





For closed loop ADCA ,

$$10 - 4(I_1 - I_2) + 2(I_2 + I_3 - I_1) - I_1 = 0$$

$$7I_1 - 6I_2 - 2I_3 = 10 \text{ -----(i)}$$

For closed loop ABCA ,

$$10 - 4I_2 - 2(I_2 + I_3) - I_1 = 0$$

$$I_1 + 6I_2 + 2I_3 = 10 \text{ -----(ii)}$$

For closed loop BCDED ,

$$5 - 2(I_2 + I_3) - 2(I_2 + I_3 - I_1) = 0$$

$$2I_1 - 4I_2 - 4I_3 = -5 \text{ -----(iii)}$$

$$\text{Current in branch AB} = I_2 = \frac{5}{8} A$$

$$\text{Current in branch AC} = I_1 = 2.5 A$$

$$\text{Current in branch BC} = I_2 + I_3 = 2.5 A$$

1/2

1/2

1/2

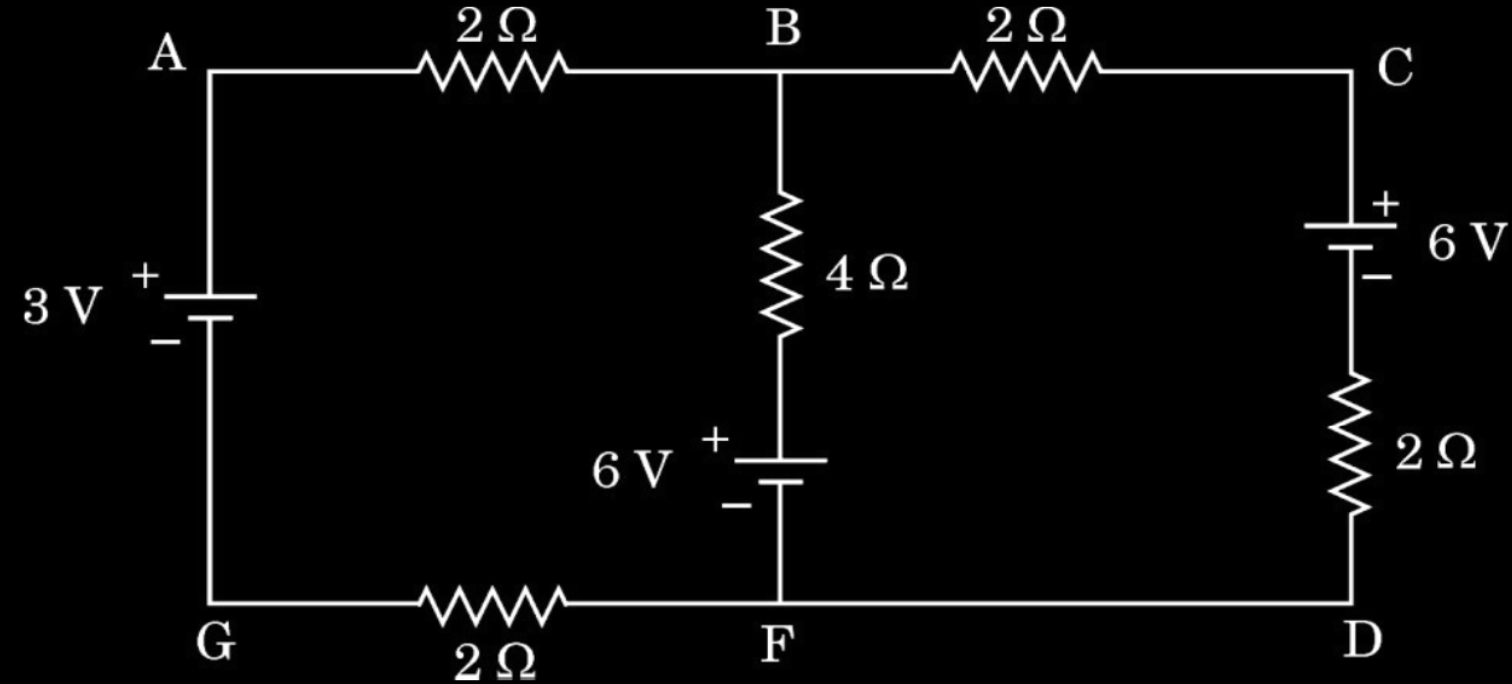
1/2

1/2

1/2

25. The figure shows a circuit with three ideal batteries. Find the magnitude and direction of currents in the branches AG, BF and CD.

3



Finding magnitude and direction of current in AG, BF and CD 1+1+1

By Kirchoff's Laws (at point B)

$$I_1 + I_2 = I_3 \quad \dots\dots(1)$$

In the closed loop AGFBA

$$3 + 2I_3 - 6 + 4I_2 + 2I_3 = 0$$

$$I_2 + I_3 = \frac{3}{4} \quad \dots\dots(2)$$

From (i)

$$2I_1 + I_2 = \frac{3}{4} \quad \dots\dots(3)$$

In closed loop BFDCB

$$-4I_2 + 6 + 2I_1 - 6 + 2I_1 = 0$$

$$I_2 - I_1 = 0$$

$$I_2 = I_1 \quad \dots\dots(4)$$

Putting in (3)

$$I_1 = \frac{1}{4} A$$

From (4)

$$I_2 = \frac{1}{4} A$$

$$\text{From (2)} \quad I_3 = \frac{1}{2} A$$

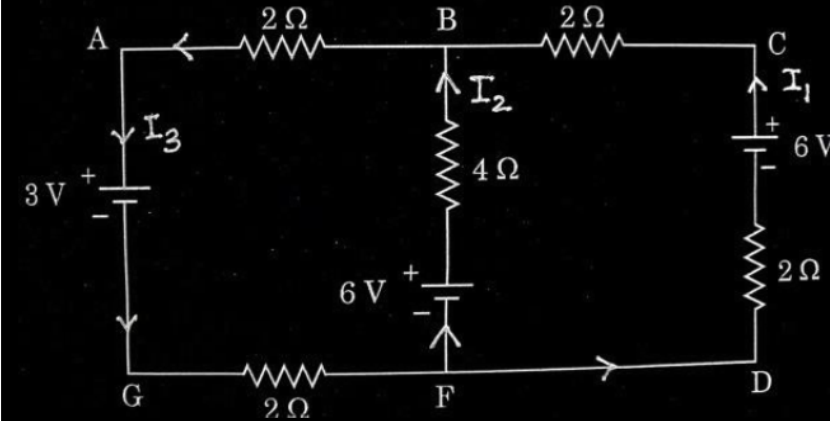
$$\frac{1}{2}$$

$$\frac{1}{2}$$

$$\frac{1}{2}$$

$$\frac{1}{2}$$

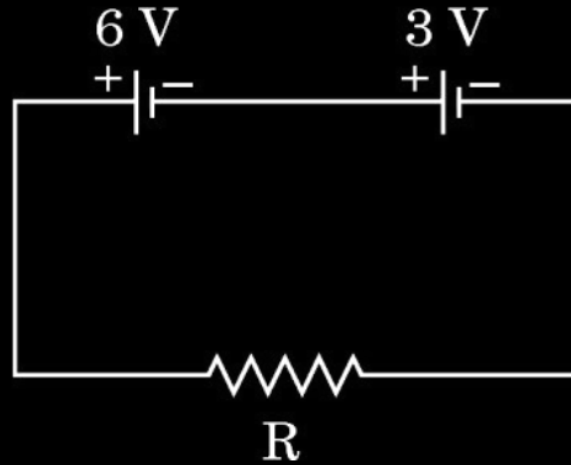
$$\frac{1}{2}$$



$$\frac{1}{2}$$

17. Two batteries of emfs 6 V and 3 V and internal resistances $0.8\ \Omega$ and $0.2\ \Omega$ respectively are connected in series to an external resistance R , as shown in figure. Find the value of R so that the potential difference across the 6 V battery be zero.

2



Finding the value of R

2

$$i = \frac{9}{R+1}$$

As potential difference across 6V is zero;

$$6 - ir = 0 \Rightarrow 6 - \left(\frac{9}{R+1} \right) (0.8) = 0$$

On solving;

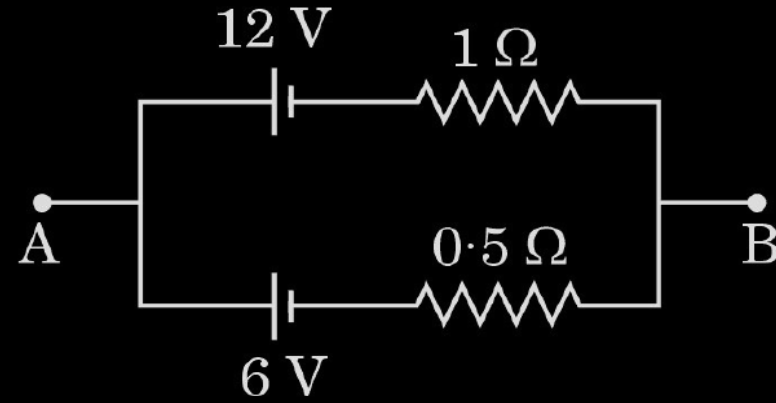
$$R = 0.2\Omega$$

$\frac{1}{2}$

$\frac{1}{2}$

1

3. Consider the circuit shown in the figure. The potential difference between points A and B is :



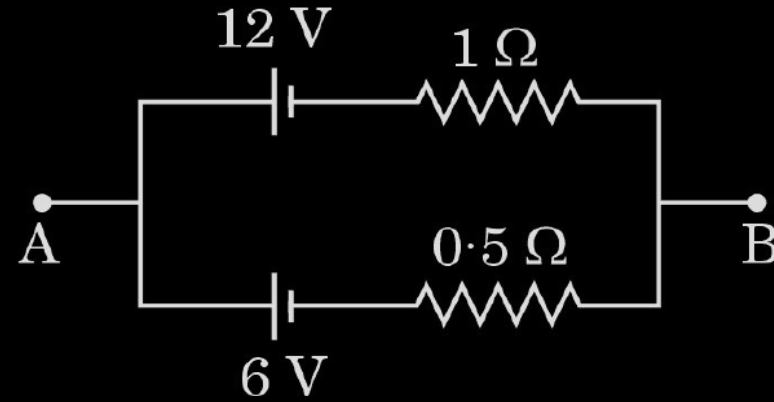
(A) 6 V

(B) 8 V

(C) 9 V

(D) 12 V

3. Consider the circuit shown in the figure. The potential difference between points A and B is :



(A) 6 V

(B) 8 V

(C) 9 V

(D) 12 V

(B) 8 V

22. Three cells A, B and C of emfs 2 V, 3 V and 5 V respectively are connected in parallel to each other. Their internal resistances are $5\ \Omega$, $5\ \Omega$ and $1\ \Omega$ respectively. Calculate the currents flowing through the cells A, B and C. 3

22. Three cells A, B and C of emfs 2 V, 3 V and 5 V respectively are connected in parallel to each other. Their internal resistances are $5\ \Omega$, $5\ \Omega$ and $1\ \Omega$ respectively. Calculate the currents flowing through the cells A, B and C. 3

In loop ABEFA,

$$5 - 3 - 5I_1 - I = 0$$

$$2 = 5I_1 + I \quad \text{-----(1)}$$

In loop CBEDC,

$$2 - 3 - 5I_1 + 5I - 5I_1 = 0$$

$$-1 = 10I_1 - 5I \quad \text{-----(2)}$$

Solving equation (1) and (2)

$$I = \frac{5}{7} \text{ A} \quad \text{in arm AF/through the cell of 5V(A)}$$

$$I_1 = \frac{9}{35} \text{ A} \quad \text{in arm BE/through the cell of 3V(B)}$$

$$I - I_1 = \frac{16}{35} \text{ A} \quad \text{in arm CD/through the cell of 2V(C)}$$

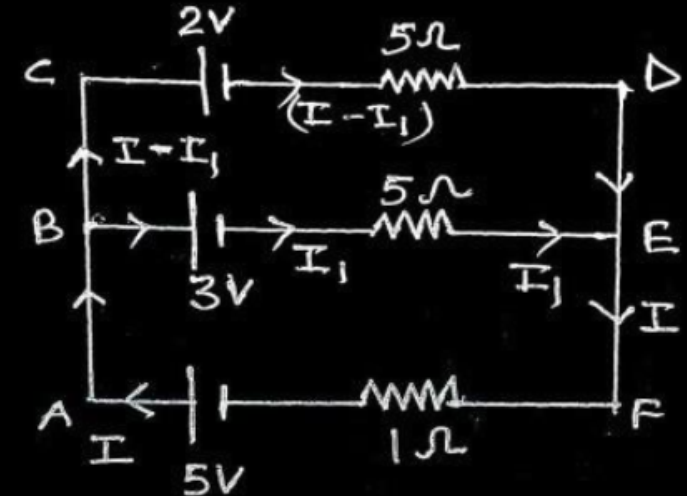
$$\frac{1}{2}$$

$$\frac{1}{2}$$

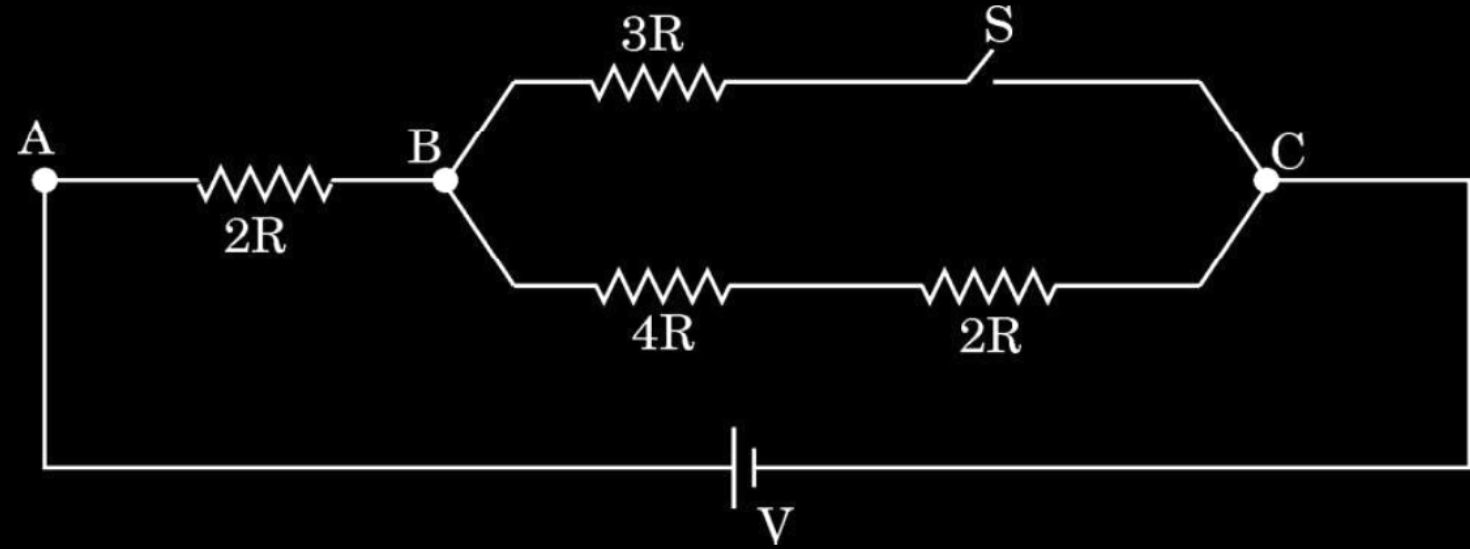
$$\frac{1}{2}$$

$$\frac{1}{2}$$

$$\frac{1}{2}$$



1. The ratio of potential difference across AB in the circuit shown for the case (i) when switch S is closed and (ii) when S is open is :



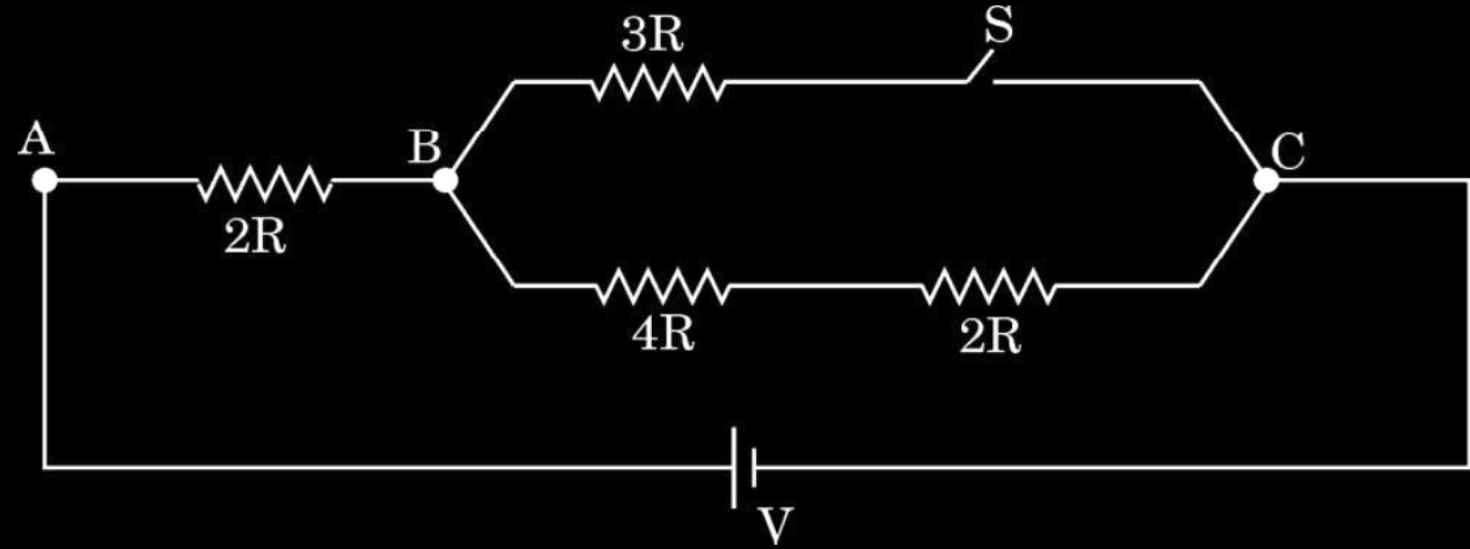
(A) $\frac{1}{4}$

(B) $\frac{1}{2}$

(C) 1

(D) 2

1. The ratio of potential difference across AB in the circuit shown for the case (i) when switch S is closed and (ii) when S is open is :



(A) $\frac{1}{4}$

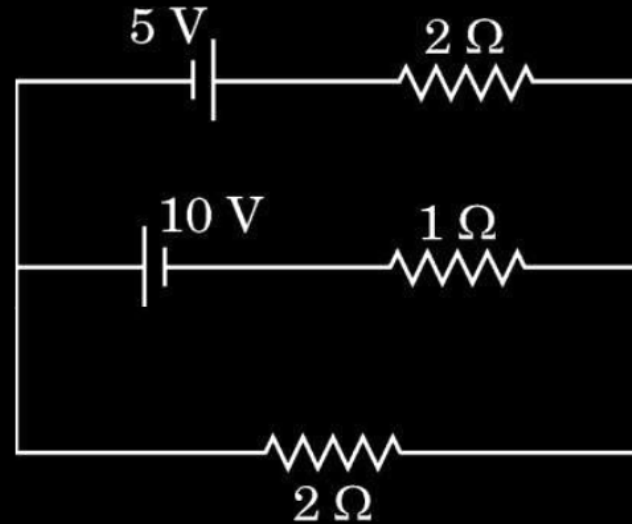
(B) $\frac{1}{2}$

(C) 1

(D) 2

26. State Kirchhoff's laws. Apply these laws to find the values of current flowing in the three branches of the given circuit.

3



Stating Kirchhoff's laws

$\frac{1}{2} + \frac{1}{2}$

Finding the values of current in all the three branches

2

Junction rule: At any junction, the sum of the currents entering the junction is equal to the sum of the currents leaving the junction.

1/2

Loop rule: the algebraic sum of changes in potential around any closed loop involving resistors and cells in the loop is zero.

1/2

In closed loop $BCDEB$ $2I_1 + 3I_2 = 10$ ----- (1)

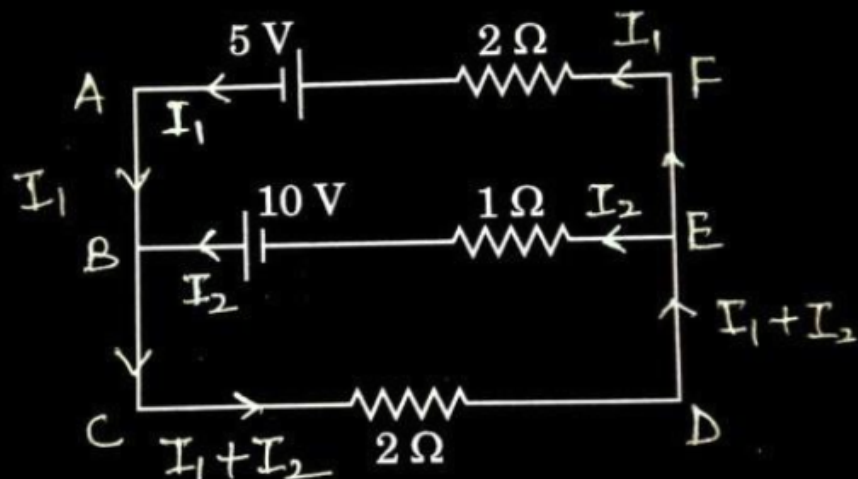
In closed loop $ABCDEFA$ $4I_1 + 2I_2 = -5$ ----- (2)

On solving eq. (1) and (2)

$$I_1 = -\frac{35}{8} \text{ A} \quad \text{in arm AF}$$

$$I_2 = \frac{25}{4} \text{ A} \quad \text{in arm BE}$$

$$I_1 + I_2 = -\frac{15}{8} \text{ A} \quad \text{in arm CD}$$



1/2

1/2

1/2

2. If R_s and R_p are the equivalent resistances of n resistors, each of value R , in series and parallel combinations respectively, then the value of $(R_s - R_p)$ is :

(A) $\left(\frac{n^2 - 1}{n^2}\right)R$

(B) $\left(\frac{n^2 + 1}{n^2 - 1}\right)R$

(C) $\left(\frac{n^2 - 1}{n}\right)R$

(D) $\frac{(n^2 + 1)R}{n^2}$

2. If R_s and R_p are the equivalent resistances of n resistors, each of value R , in series and parallel combinations respectively, then the value of $(R_s - R_p)$ is :

(A) $\left(\frac{n^2 - 1}{n^2}\right)R$

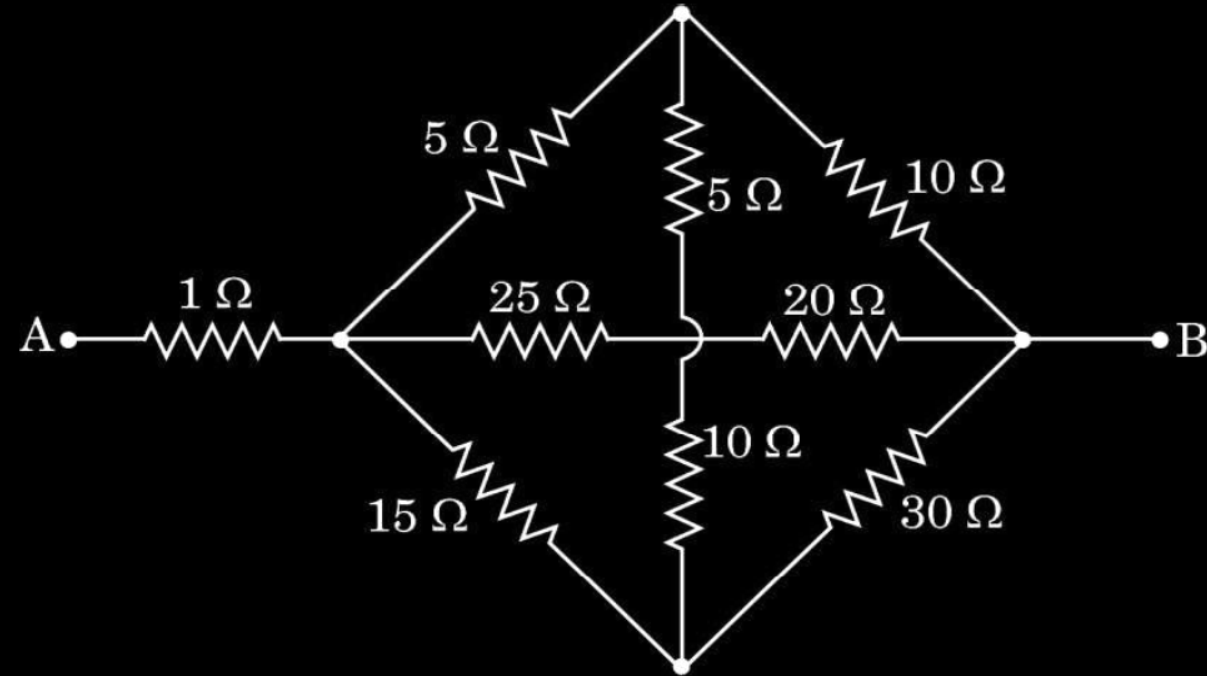
(B) $\left(\frac{n^2 + 1}{n^2 - 1}\right)R$

(C) $\left(\frac{n^2 - 1}{n}\right)R$

(D) $\frac{(n^2 + 1)R}{n^2}$

17. Find the equivalent resistance between points A and B for the network shown in the figure.

2



Resistance between points C and B

$$\frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}$$

$$\frac{1}{R} = \frac{1}{15} + \frac{1}{45} + \frac{1}{45}$$

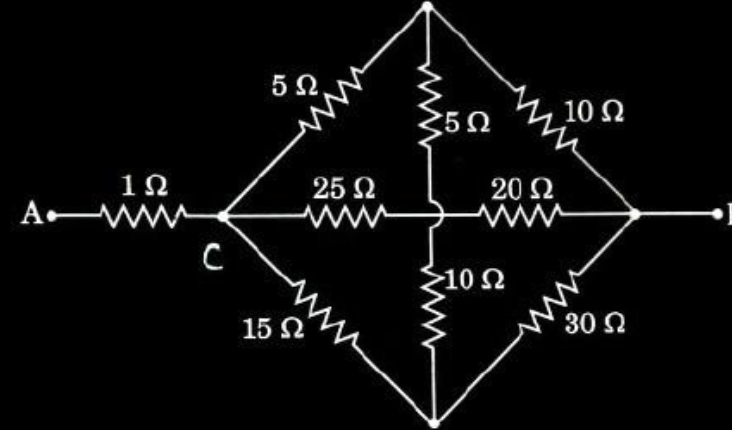
$$R = 9 \, \Omega$$

Equivalent resistance between points A and B

$$R_{eq} = R_1 + R_2$$

$$R_{eq} = 1 + 9$$

$$= 10 \, \Omega$$



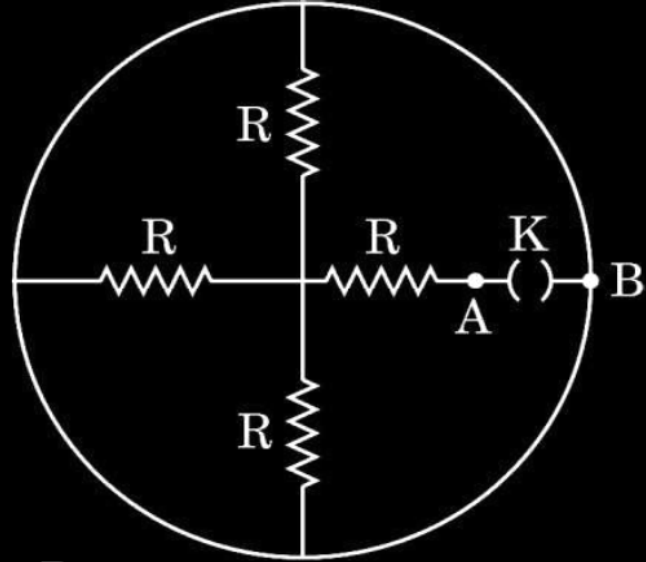
1/2

1/2

1/2

1/2

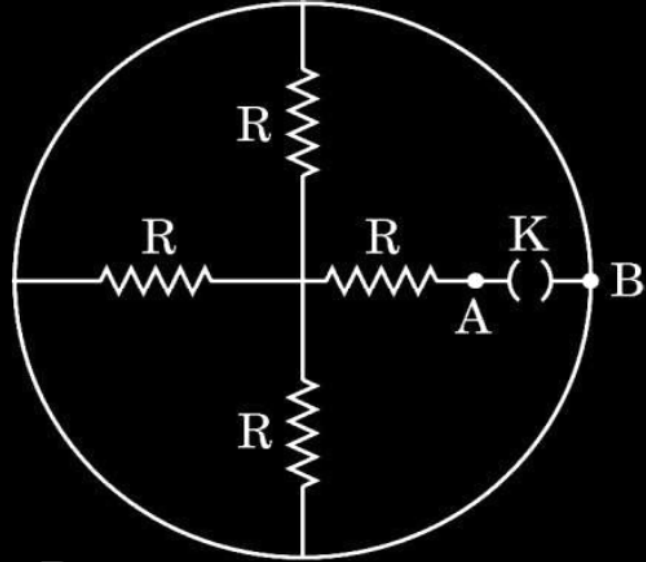
5. Four resistors, each of resistance R and a key K are connected as shown in the figure. The equivalent resistance between points A and B when key K is open, will be :



- (A) $4R$
(C) $\frac{R}{4}$

- (B) ∞
(D) $\frac{4R}{3}$

5. Four resistors, each of resistance R and a key K are connected as shown in the figure. The equivalent resistance between points A and B when key K is open, will be :



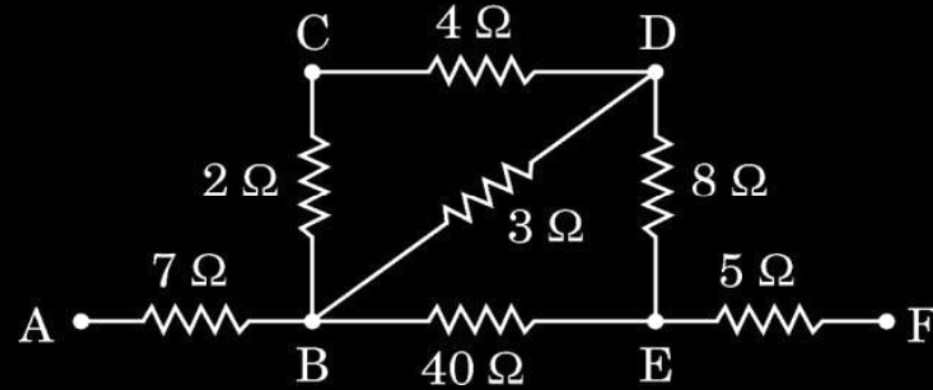
- (A) $4R$
(C) $\frac{R}{4}$

- (B) ∞

- (D) $\frac{4R}{3}$

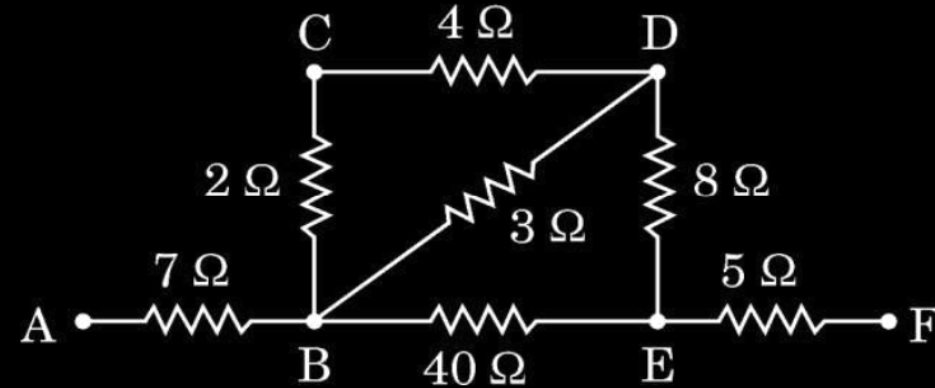
17. Find the effective resistance of the network of resistors between points A and F as shown in the figure.

2



17. Find the effective resistance of the network of resistors between points A and F as shown in the figure.

2



$$R_{BCD} = 2 + 4 = 6\Omega$$

$$R_{BD} = \frac{3 \times 6}{6 + 3}$$

$$= 2\Omega$$

$$R_{BDE} = 2 + 8$$

$$= 10\Omega$$

$$R_{BE} = \frac{40 \times 10}{40 + 10}$$

$$= 8\Omega$$

$$R_{AF} = 7 + 8 + 5$$

$$= 20\Omega$$

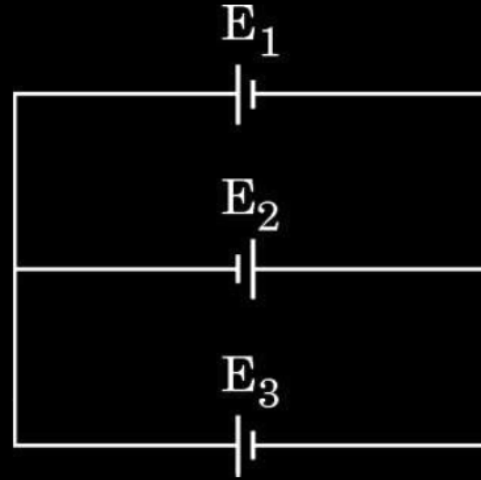
$\frac{1}{2}$

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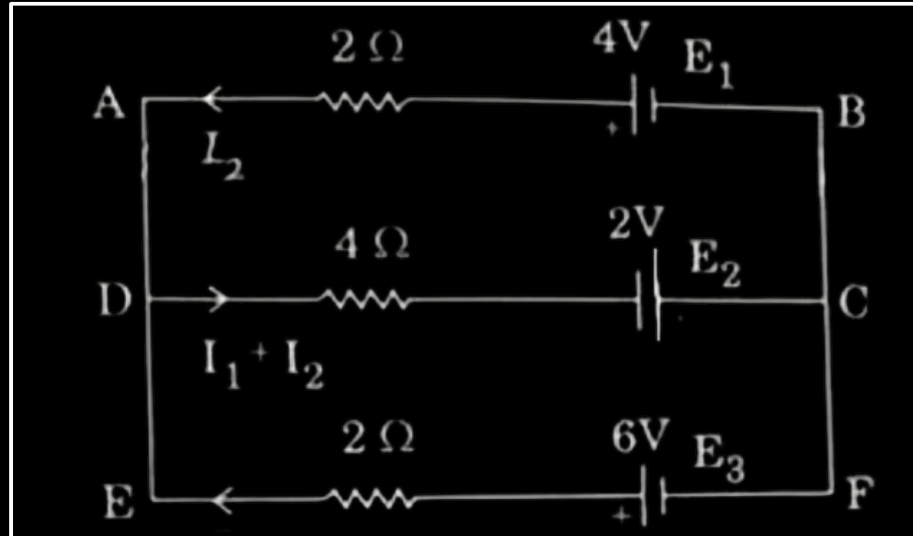
$\frac{1}{2}$

$\frac{1}{2}$

- 32.** (a) (i) Three batteries E_1 , E_2 and E_3 of emfs and internal resistances (4 V, $2\ \Omega$), (2 V, $4\ \Omega$) and (6 V, $2\ \Omega$) respectively are connected as shown in the figure. Find the values of the currents passing through batteries E_1 , E_2 and E_3 .



- | | | |
|------|---|---|
| (i) | Finding current through batteries E_1 , E_2 and E_3 | 3 |
| (ii) | Finding effective resistance | 2 |



i)

In closed loop ABCD, using Kirchhoff's loop law

$$4I_1 + 6I_2 = 6 \dots\dots\dots(1)$$

Similarly In closed loop CDFE

$$6I_1 + 4I_2 = 8 \dots\dots\dots(2)$$

Solving eqn. (1) and (2)

$$I_2 = \frac{1}{5} \text{A}$$

$$I_1 = \frac{6}{5} \text{A}$$

$$I_1 + I_2 = \frac{7}{5} \text{A}$$

$\frac{1}{2}$

$\frac{1}{2}$

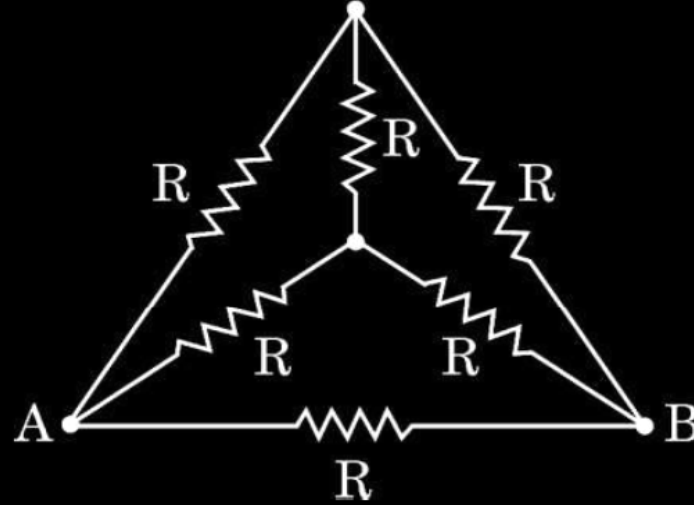
$\frac{1}{2}$

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$\frac{1}{2}$

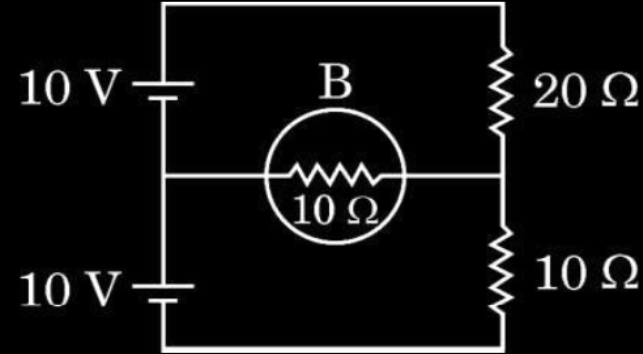
- (ii) The ends of six wires, each of resistance R ($= 10\ \Omega$) are joined as shown in the figure. The points A and B of the arrangement are connected in a circuit. Find the value of the effective resistance offered by it to the circuit.

5



OR

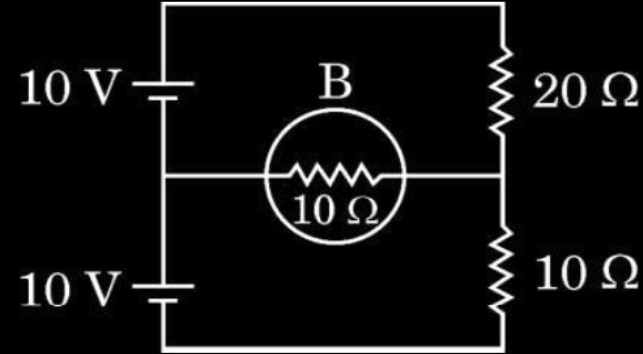
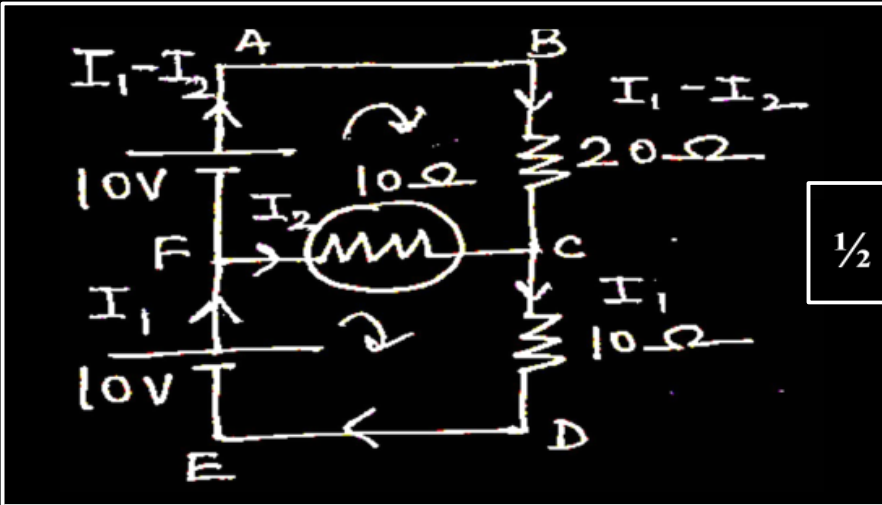
- (b) Two cells of emf 10 V each, two resistors of $20\ \Omega$ and $10\ \Omega$ and a bulb B of $10\ \Omega$ resistance are connected together as shown in the figure. Find the current that flows through the bulb.



OR

- (b) Two cells of emf 10 V each, two resistors of $20\ \Omega$ and $10\ \Omega$ and a bulb B of $10\ \Omega$ resistance are connected together as shown in the figure. Find the current that flows through the bulb.

2



By applying Kirchhoff's loop rule to closed loops ABCFA and FCDEF

$$2I_1 - 3I_2 = 1 \text{ ---- (1)}$$

$$I_1 + I_2 = 1 \text{ ---- (2)}$$

On solving,

Current through the bulb,

$$I_2 = \frac{1}{5} \text{ A}$$

$$\frac{1}{2}$$

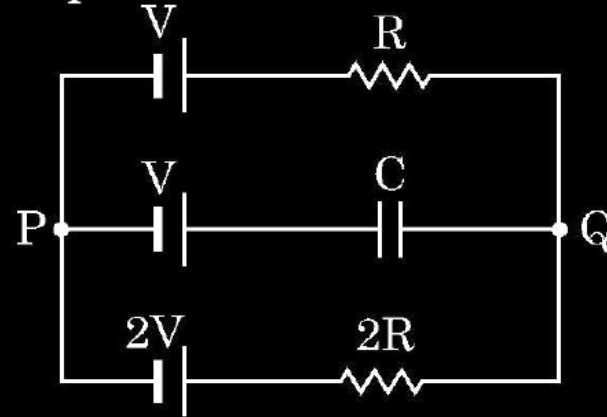
$$\frac{1}{2}$$

$$\frac{1}{2}$$

21. (a) Two wires of the same material and the same radius have their lengths in the ratio $2 : 3$. They are connected in parallel to a battery which supplies a current of 15 A . Find the current through the wires. 2

OR

- (b) In the circuit three ideal cells of e.m.f. V , V and $2V$ are connected to a resistor of resistance R , a capacitor of capacitance C and another resistor of resistance $2R$ as shown in figure. In the steady state find (i) the potential difference between P and Q and (ii) potential difference across capacitor C .



(b)

Finding the potential difference

(i) between P and Q

1

(ii) across capacitor C

1

In steady state,

$$2V - V = i(2R + R)$$

$$i = \frac{V}{3R}$$

$$(i) \quad V_P - V_Q = -V - iR$$

$$= -V - \frac{V}{3}$$

$$V_P - V_Q = -\frac{4V}{3}$$

1

$$(ii) \quad V_P - V_Q = -V + V_C$$

$$-\frac{4V}{3} = -V + V_C$$

1

$$V_C = -\frac{V}{3}$$

2. A uniform wire of resistance $2R$ is bent in the form of a circle. The effective resistance between the ends of any diameter of the circle is :

- (a) $2R$ (b) R (c) $\frac{R}{2}$ (d) $\frac{R}{4}$

2. A uniform wire of resistance $2R$ is bent in the form of a circle. The effective resistance between the ends of any diameter of the circle is :

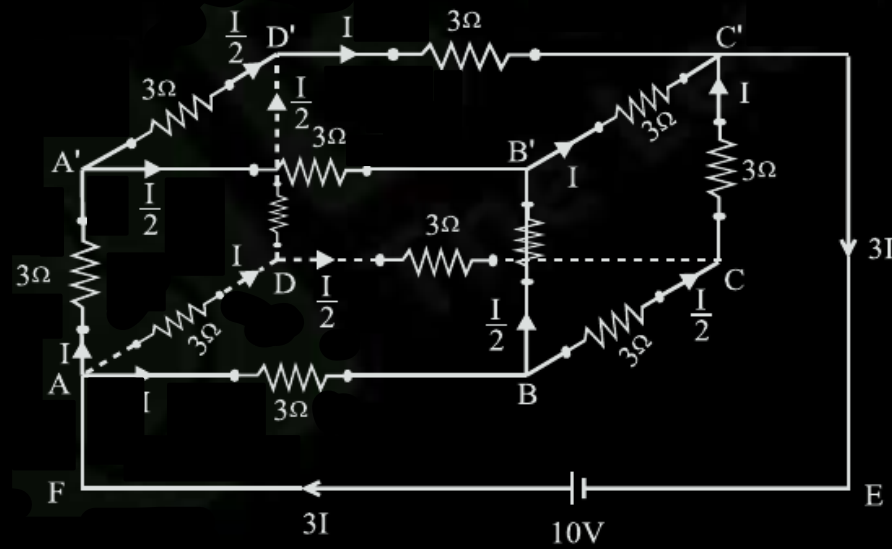
(a) $2R$

(b) R

(c) $\frac{R}{2}$

(d) $\frac{R}{4}$

Twelve wires each having a resistance of $3\ \Omega$ are connected to form a cubical network. A battery of $10\ \text{V}$ and negligible internal resistance is connected across the diagonally opposite corners of this network. Determine its equivalent resistance and the current along each edge of the cube.



$\frac{1}{2}$

Applying loop rule to ABCC'EFA

$$3I + 3\frac{I}{2} + 3I - 10 = 0$$

$\frac{1}{2}$

$$\frac{15}{2} I = 10$$

$$I = \frac{2 \times 10}{15} = \frac{20}{15} \text{ A} = \frac{4}{3} \text{ A}$$

$\frac{1}{2}$

$$R_{\text{tot}} = \frac{V}{3I} = \frac{10 \times 15}{3 \times 20} = 2.5\Omega$$

$\frac{1}{2}$

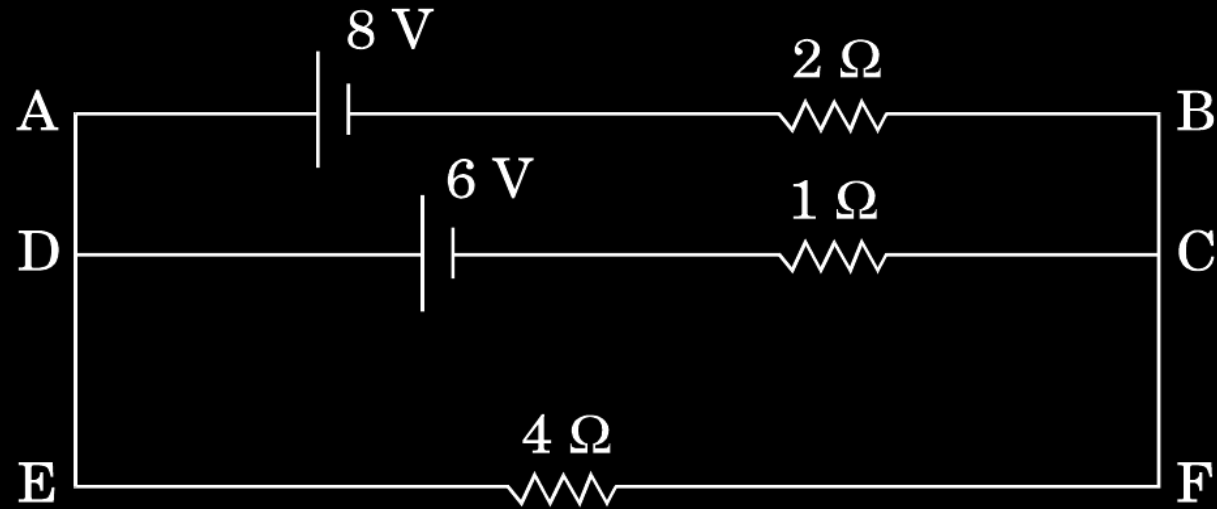
$$\text{Current} = I_{AB} (= I_{AA'} = I_{AD} = I_{D'C'} = I_{B'C'} = I_{CC'}) = \frac{4}{3} \text{ A}$$

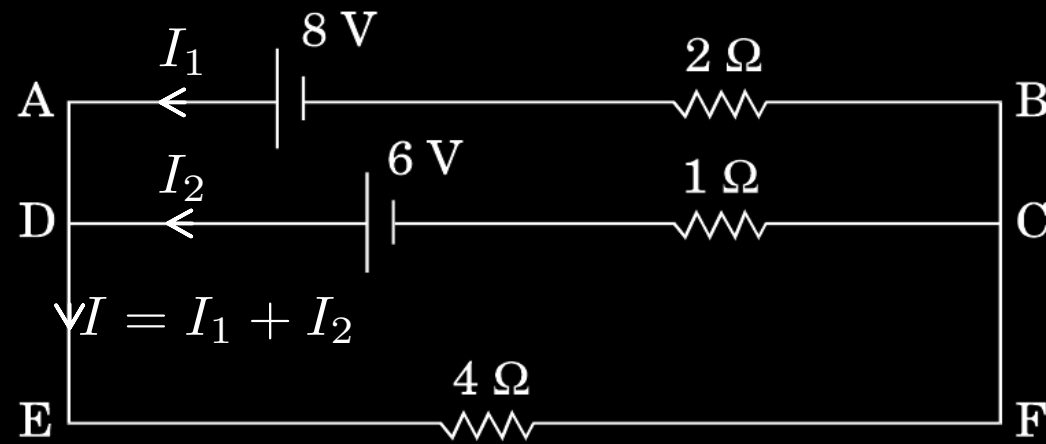
$\frac{1}{2}$

$$I_{DD'} (= I_{A'B'} = I_{A'D'} = I_{DC} = I_{BC} = I_{BB'}) = \frac{2}{3} \text{ A}$$

$\frac{1}{2}$

- (c) Calculate the potential difference across the $4\ \Omega$ resistor in the given electrical circuit, using Kirchhoff's rules.





$$I = I_1 + I_2 \quad (1)$$

In loop ABCDA

$$-8 + 2I_1 - 1 \times I_2 + 6 = 0 \quad (2)$$

In loop DEFC

$$-4I - 1 \times I_2 + 6 = 0$$

$$4I + I_2 = 6$$

$$4(I_1 + I_2) + I_2 = 6$$

$$4I_1 + 5I_2 = 6 \quad (3)$$

From equations (1) and (2) we get

$$I_1 = \frac{8}{7} \text{ A}, \quad I_2 = \frac{2}{7} \text{ A}, \quad I = \frac{10}{7}$$

Potential difference across resistor 4Ω is:

$$V = \frac{10}{7} \times 4 = \frac{40}{7} \text{ volt}$$

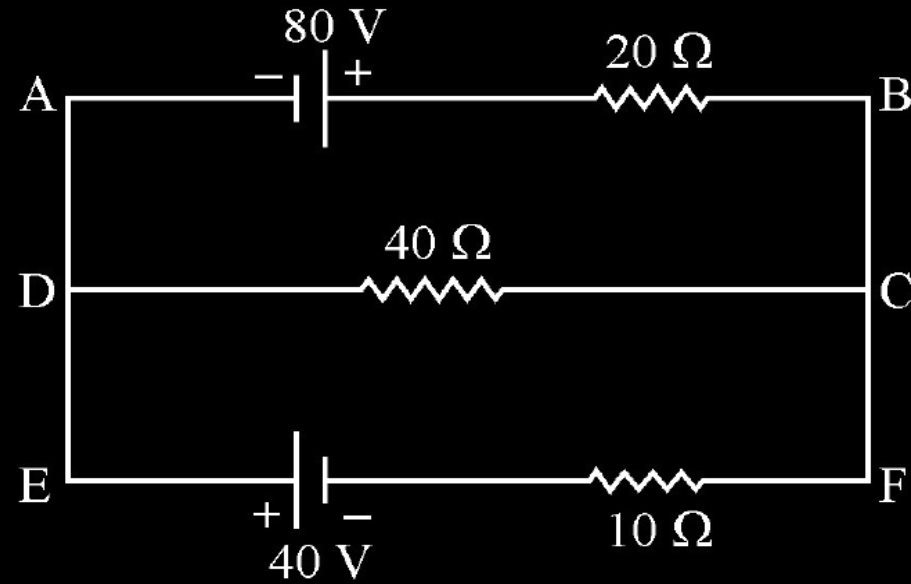
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1

$\frac{1}{2}$

$\frac{1}{2}$

Using Kirchhoff's rules, calculate the current through the $40\ \Omega$ and $20\ \Omega$ resistors in the following circuit :



In loop ABCFA

$$+80 - 20 I_2 + 40 I_1 = 0$$

$$4 = I_2 - 2 I_1$$

In loop FCDEA

$$-40 I_1 - 10(I_1 + I_2) + 40 = 0$$

$$-50 I_1 - 10 I_2 + 40 = 0$$

$$5 I_1 + I_2 = 4$$

Solving these two equations

$$I_1 = 0A$$

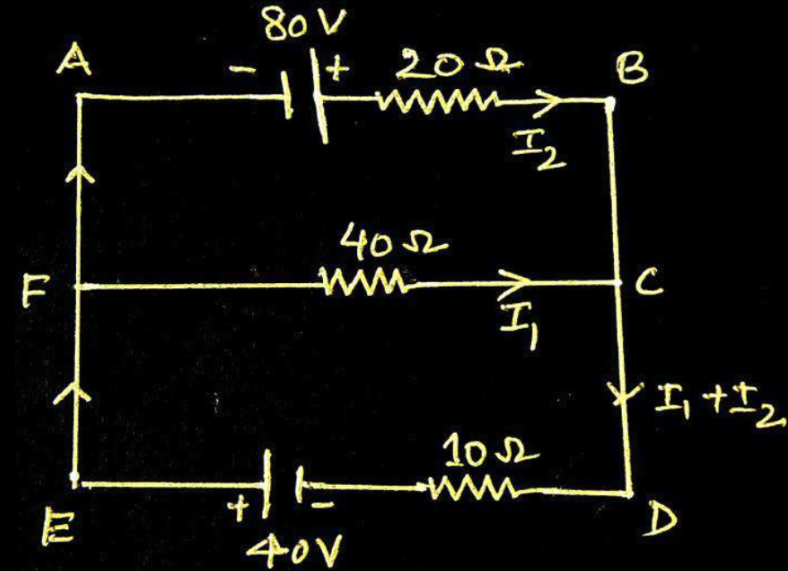
$$\& I_2 = 4A$$

1

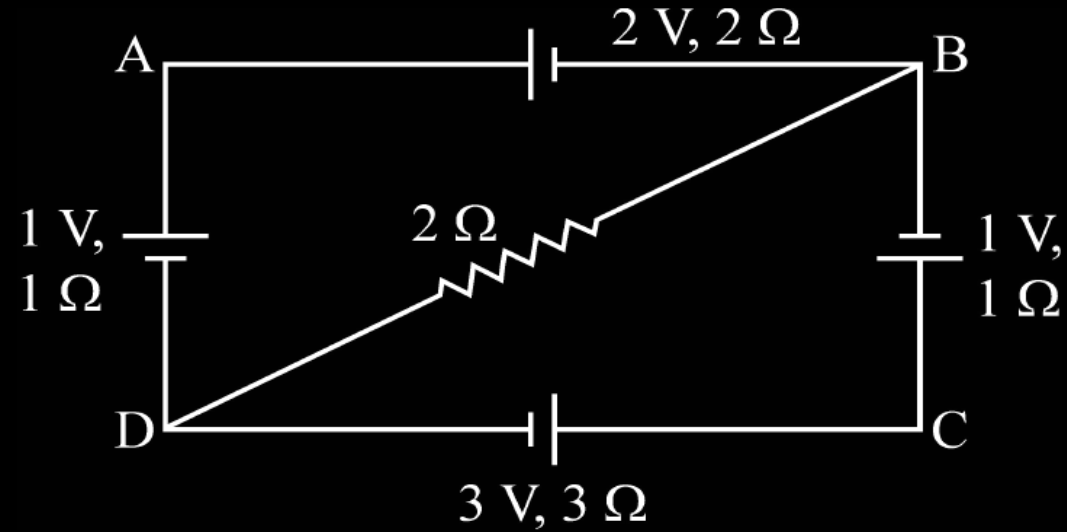
1

$\frac{1}{2}$

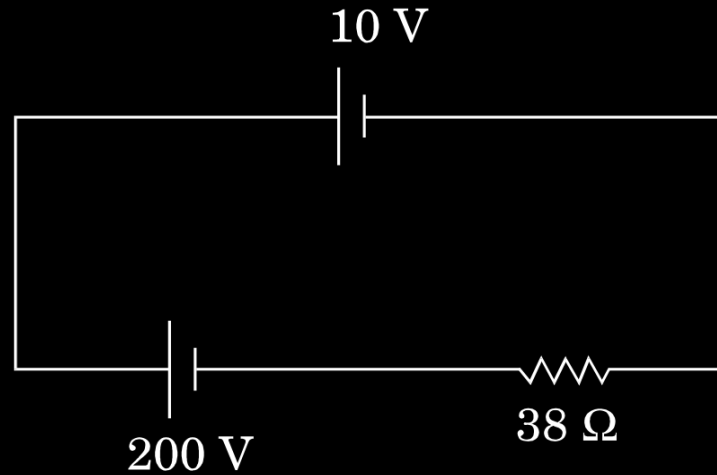
$\frac{1}{2}$



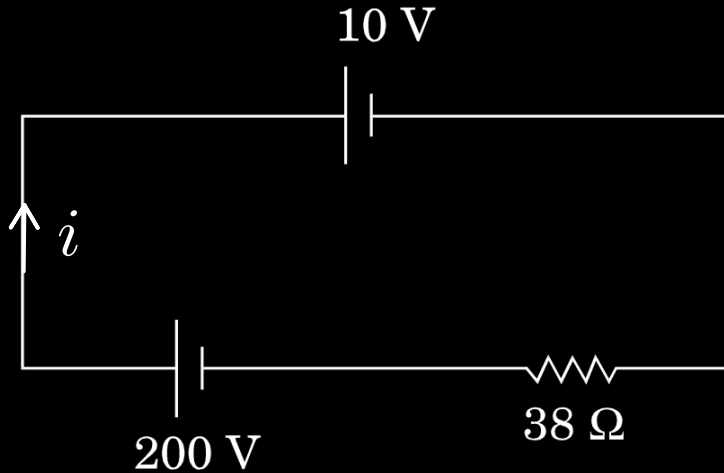
Using Kirchhoff's rules, calculate the potential difference between B and D in the circuit diagram as shown in the figure.



A 10 V cell of negligible internal resistance is connected in parallel across a battery of emf 200 V and internal resistance $38\ \Omega$ as shown in the figure. Find the value of current in the circuit.



A 10 V cell of negligible internal resistance is connected in parallel across a battery of emf 200 V and internal resistance $38\ \Omega$ as shown in the figure. Find the value of current in the circuit.



By Kirchhoff's law, we have, for the loop ABCD,

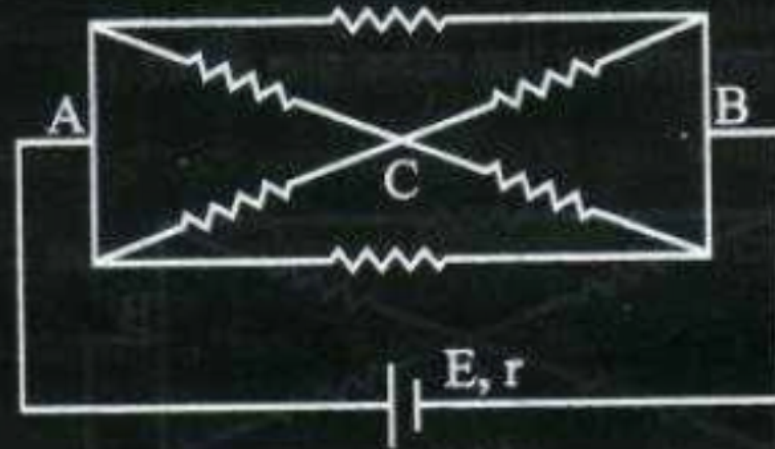
$$+200 - 38i - 10 = 0$$

$$\therefore i = \frac{190}{38} \text{ A} = 5 \text{ A}$$

1

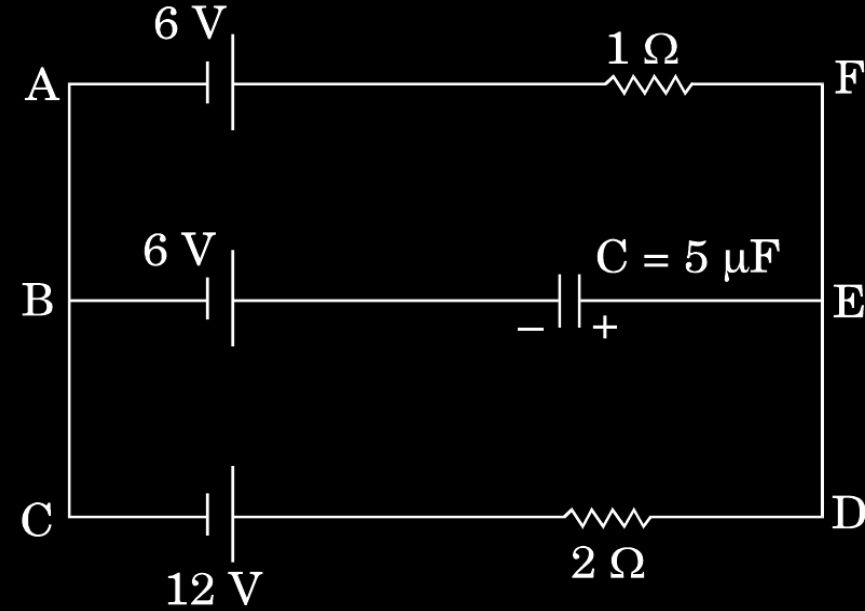
1

- (i) State the two Kirchhoff's laws. Explain briefly how these rules are justified.
- (ii) The current is drawn from a cell of emf E and internal resistance r connected to the network of resistors each of resistance r as shown in the figure. Obtain the expression for (i) the current draw from the cell and (ii) the power consumed in the network.

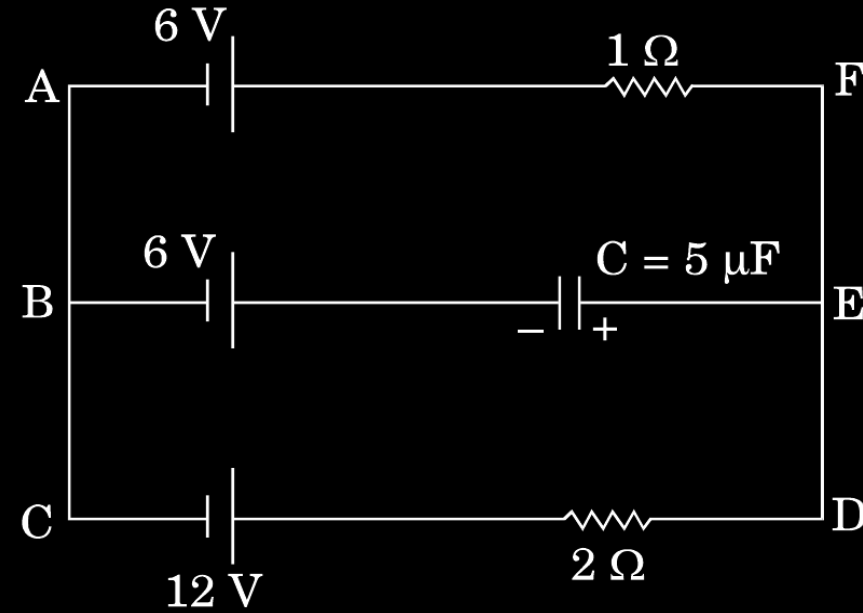


(i) Junction Rule: At any Junction, the sum of currents, entering the junction, is equal to the sum of currents leaving the junction. Loop Rule: The Algebraic sum, of changes in potential, around any closed loop involving resistors and cells, in the loop is zero. $\sum(\Delta V) = 0$ Justification: The first law is in accord with the law of conservation of charge. The Second law is in accord with the law of conservation of energy.	1 1 1/2 1/2
(ii) Equivalent resistance of the loop $R = r/3$ Hence current drawn from the cell $I = \frac{E}{r/3 + r} = \frac{3E}{4r}$ Power consumed $P = I^2 (r/3)$ $= \frac{9E^2}{16r^2} \times \frac{4r}{3} = \frac{3E^2}{4r}$ [Note: Award the last 1 1/2 marks for this part, if the calculations, for these parts, are done by using (any other) value of equivalent resistance obtained by the student.)	1/2 1/2 1/2 1/2 1/2

In the given circuit, with steady current, calculate the potential difference across the capacitor and the charge stored in it.



In the given circuit, with steady current, calculate the potential difference across the capacitor and the charge stored in it.



In loop ACDF A

$$I = \frac{12-6}{(1+2)} = 2A$$

$$V_{AF} = V_{BE}$$

$$\Rightarrow 6 + 2 = 6 + V_c$$

$$\Rightarrow V_c = 2V$$

$$\text{Charge } Q = CV_c = 5\mu F \times 2V = 10\mu C$$

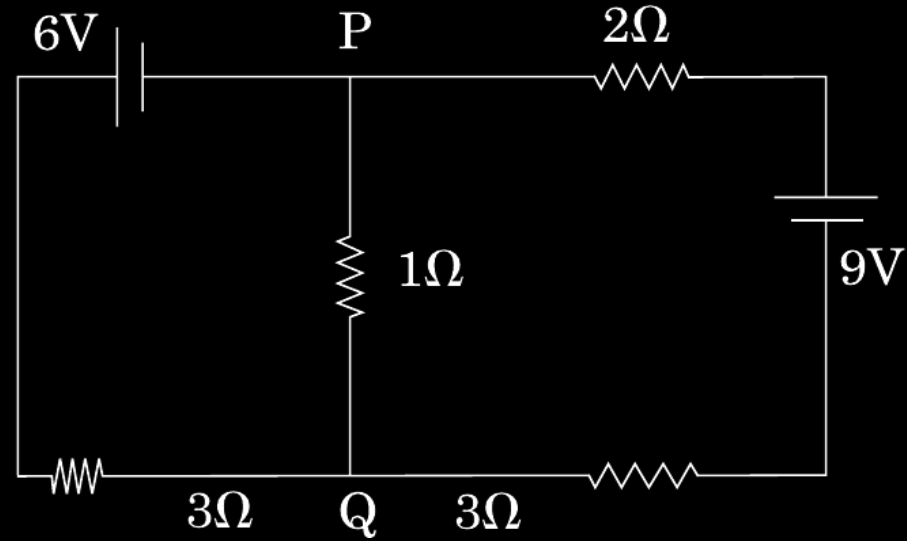
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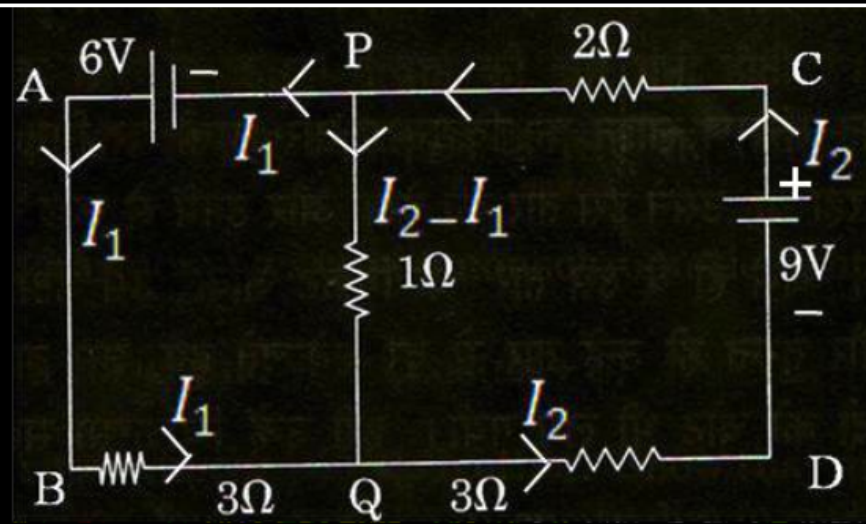
$\frac{1}{2}$

$\frac{1}{2}$

1

- (i) Find the magnitude and direction of current in $1\ \Omega$ resistor in the given circuit.





For the mesh APQBA

$$-6 - 1(I_2 - I_1) + 3I_1 = 0$$

$$\text{Or } -I_2 + 4I_1 = 6 \dots\dots\dots(1)$$

For the mesh PCDQP

$$2I_2 - 9 + 3I_2 + 1(I_2 - I_1) = 0$$

$$\text{Or } 6I_2 - I_1 = 9 \dots\dots\dots(2)$$

Solving (1) and (2) , we get

$$I_1 = \frac{45}{23} \text{ A}$$

$$I_2 = \frac{42}{23} \text{ A}$$

$$\therefore \text{Current through the } 1\Omega \text{ resistor} = \frac{-3}{23} \text{ A}$$

1

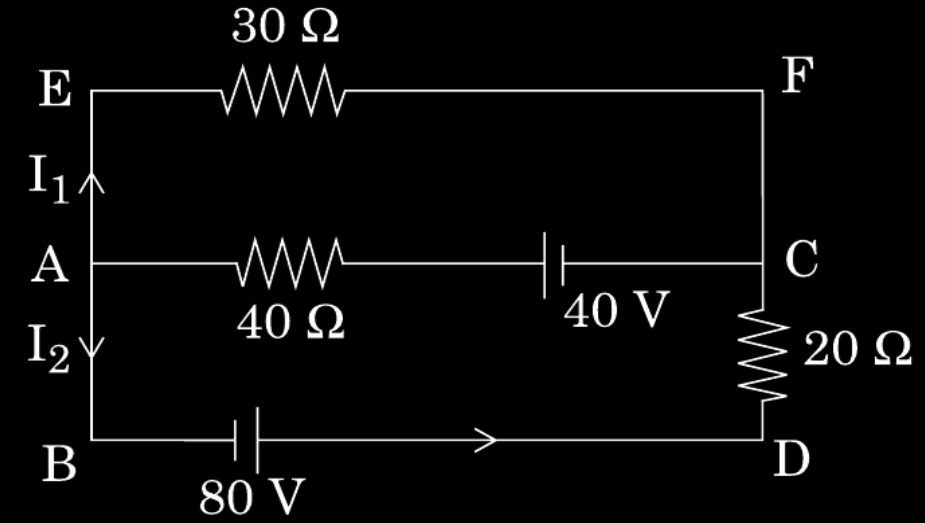
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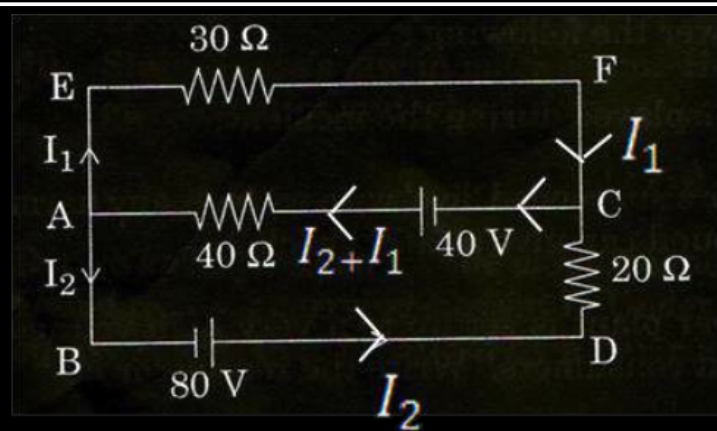
$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

- (a) Use Kirchhoff's rules, calculate the current in the arm AC of the given circuit.





For the mesh EFCAE

$$-30I_1 + 40 - 40(I_1 + I_2) = 0$$

$$\text{Or } -7I_1 - 4I_2 = -4$$

$$\text{Or } 7I_1 + 4I_2 = 4 \dots\dots\dots(1)$$

For the mesh ACDBA

$$40(I_1 + I_2) - 40 + 20I_2 - 80 = 0$$

$$\text{Or } 40I_1 + 60I_2 - 120 = 0$$

$$\text{Or } 2I_1 + 3I_2 = 6 \dots\dots\dots(2)$$

Solving (1) and (2), we get

$$I_1 = \frac{-12}{13} A$$

$$I_2 = \frac{34}{13} A$$

$$\therefore \text{Current through arm AC} = I_1 + I_2$$

$$= \frac{22}{13} A$$

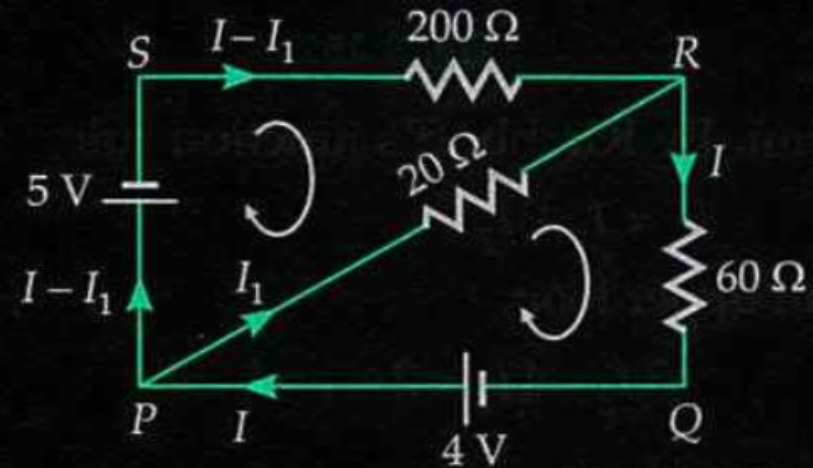
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1

1

Example 137. Network PQRS (Fig. 3.148) is made as under : PQ has a battery of 4 V and negligible resistance with positive terminal connected to P, QR has a resistance of $60\ \Omega$. PS has a battery of 5 V and negligible resistance with positive terminal connected to P, RS has a resistance of $200\ \Omega$. If a milliammeter, of $20\ \Omega$ resistance is connected between P and R, calculate the reading of the milliammeter.

[NCERT]



Solution. Applying Kirchhoff's second law to the loop $PRQP$, we get

$$20I_1 + 60I = 4 \quad \dots(i)$$

Similarly, from the loop $PSRP$, we get

$$200(I - I_1) - 20I_1 = -5$$

or $40I - 44I_1 = -1 \quad \dots(ii)$

Multiplying (i) by 2 and (ii) by 3, we get

$$120I + 40I_1 = 8 \quad \dots(iii)$$

and $120I - 132I_1 = -3 \quad \dots(iv)$

Subtracting (iv) from (iii), we get

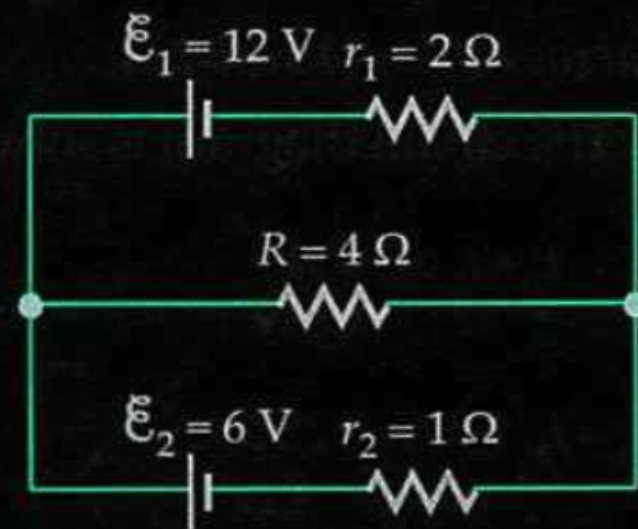
$$172I_1 = 11$$

or $I_1 = \frac{11}{172} = 0.064 \text{ A}$

Thus the milliammeter of 20Ω will read 0.064 A .

Example 139. Find the potential difference across each cell and the rate of energy dissipation in R . [Fig. 3.150(a)].

[CBSE SP 11]



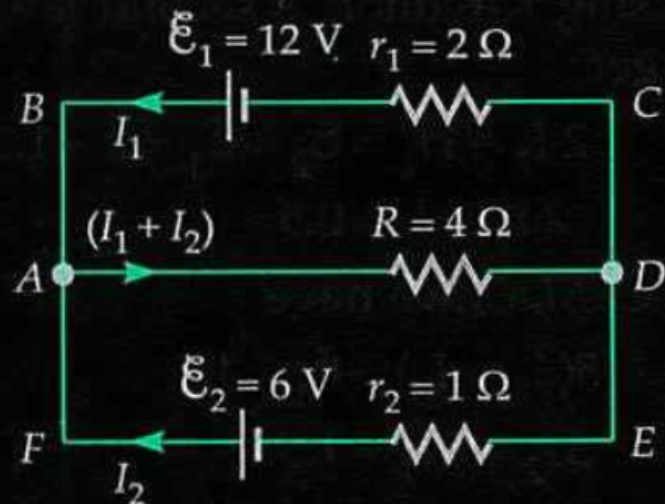
Solution. Applying Kirchhoff's laws,

For closed loop ADCBA

$$12 = 4(I_1 + I_2) + 2I_1 = 6I_1 + 4I_2 \quad \dots(i)$$

For closed loop ADEFA,

$$6 = 4(I_1 + I_2) + I_1 = 5I_1 + 4I_2 \quad \dots(ii)$$



Solving (i) and (ii), we get

$$I_1 = \frac{18}{7} \text{ A} \quad \text{and} \quad I_2 = -\frac{6}{7} \text{ A}$$

P.D. across $R = V$

$$= (I_1 + I_2) R$$

$$= \left(\frac{18-6}{7} \right) \times 4 \text{ volt} = \frac{48}{7} \text{ volt}$$

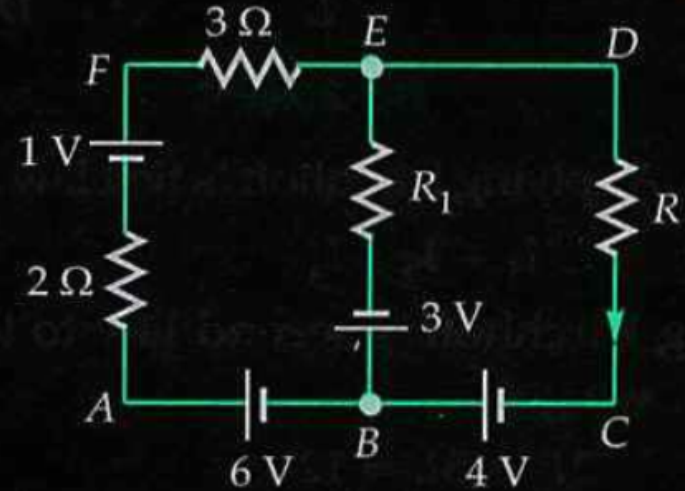
$$\text{P.D. across each cell} = \text{P.D. across } R = \frac{48}{7} \text{ V}$$

Energy dissipated in $R = 4\Omega$ resistor

$$= (I_1 + I_2)^2 R = \left(\frac{12}{7} \right)^2 \times 4 \text{ J}$$

$$= \frac{576}{49} \text{ J} = 11.75 \text{ J.}$$

Example 143. Use Kirchhoff's rules to determine the potential difference between the points A and D when no current flows in the arm BE of the electric network shown in Fig. 3.154(a). [CBSE OD 15]



Example 143. Use Kirchhoff's rules to determine the potential difference between the points A and D when no current flows in the arm BE of the electric network shown in Fig. 3.154(a). [CBSE OD 15]

Solution. No current flows through the arm BE.
Let I be the current along the outer loop as shown in Fig. 3.154(b).

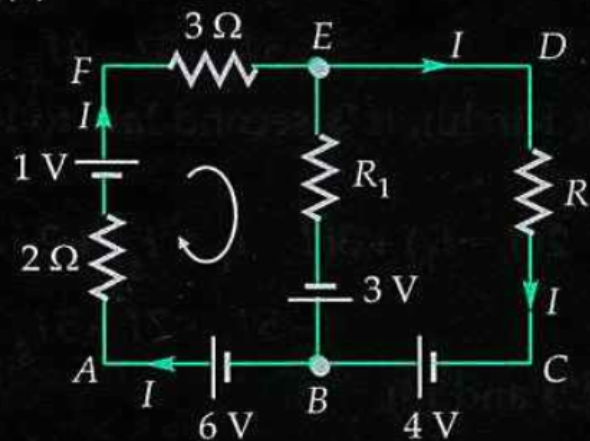


Fig. 3.154 (b)

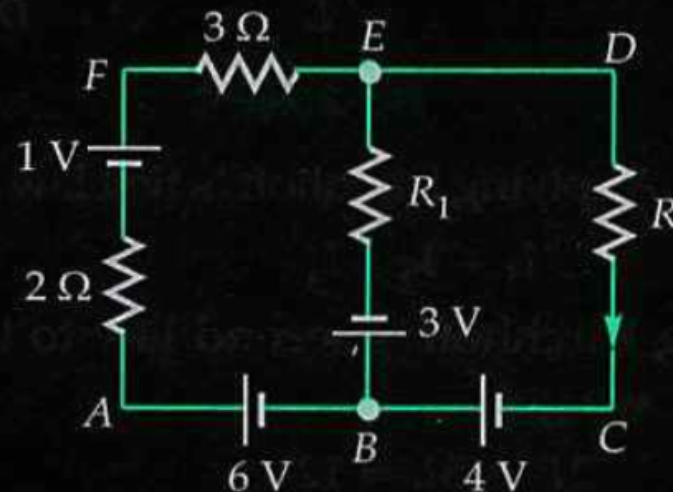
Applying Kirchhoff's loop rule to the loop AFEBA,

$$(2 + 3)I + R_1 \times 0 = 1 + 3 + 6$$

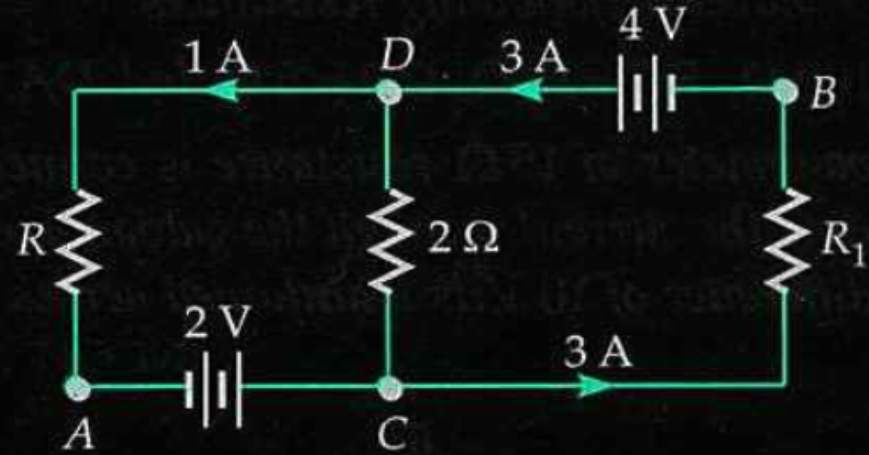
$$\therefore I = 2 \text{ A}$$

From A to D along AFD,

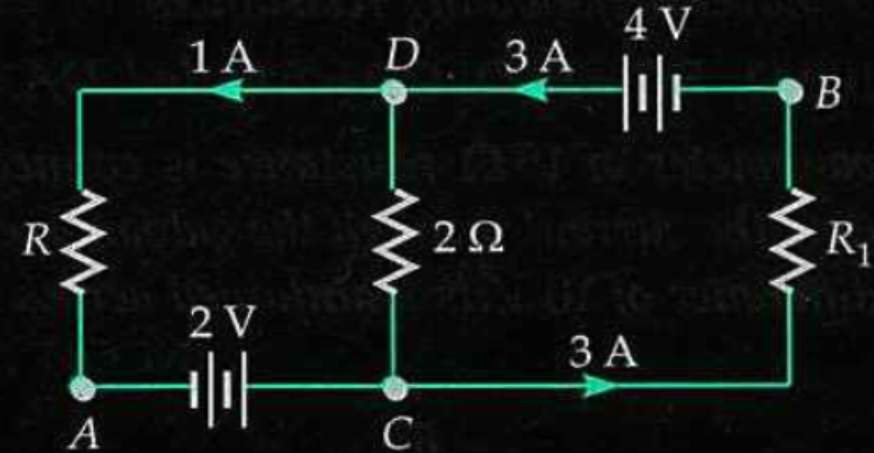
$$V_{AD} = 2 \times 2 - 1 + 3 \times 2 = 9 \text{ V.}$$



Example 144. In the circuit Fig. 3.155, assuming point A to be at zero potential, use Kirchhoff's rules to determine the potential at point B.



Example 144. In the circuit Fig. 3.155, assuming point A to be at zero potential, use Kirchhoff's rules to determine the potential at point B.



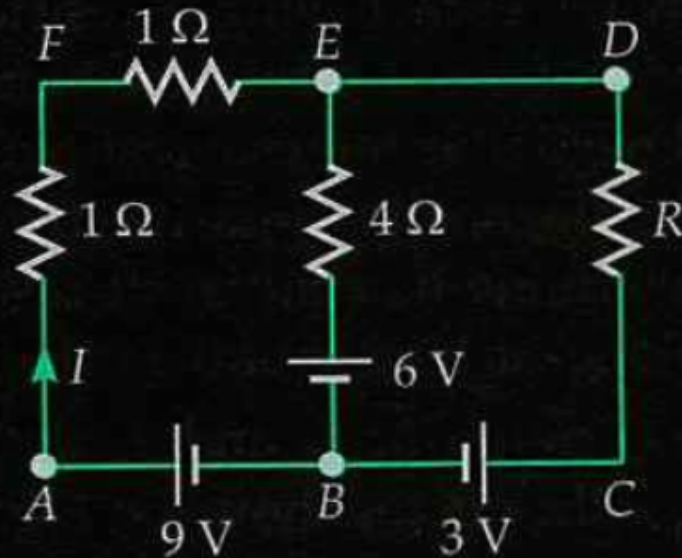
Solution. From the loop $BDCR_1B$, we get

$$2 \times 2 + 3 R_1 = 4 \quad \text{or} \quad R_1 = 0$$

$$\therefore V_B = V_C = 2 + V_A = 2 + 0 = 2 \text{ V.}$$

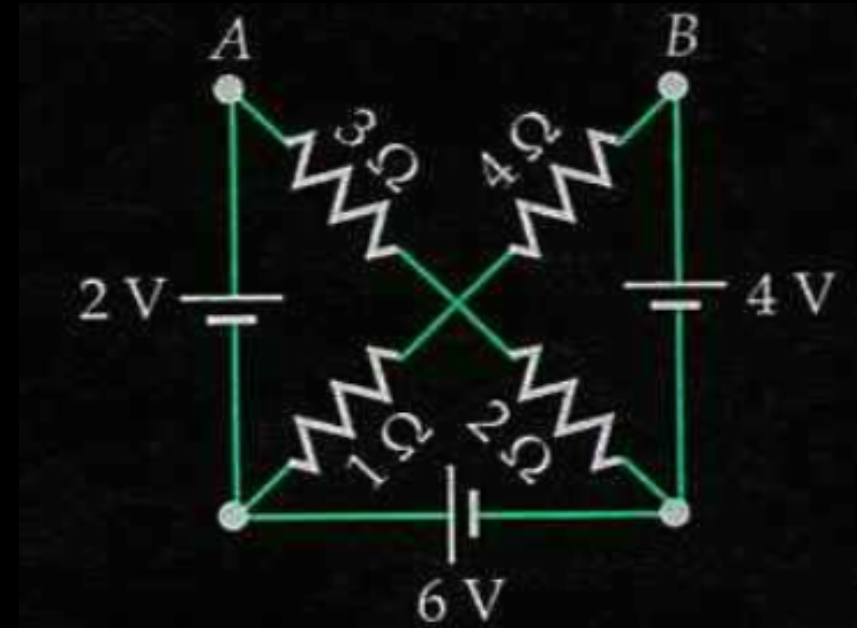
9. Using Kirchhoff's rules, determine the value of unknown resistance R in the circuit shown in Fig. 3.180 so that no current flows through 4Ω resistance. Also find the potential difference between A and D . [CBSE D 12]

(Ans. 3 V)



11. In the network shown in Fig. 3.182, (i) calculate the current of the 6 V battery and (ii) determine the potential difference between the points A and B.

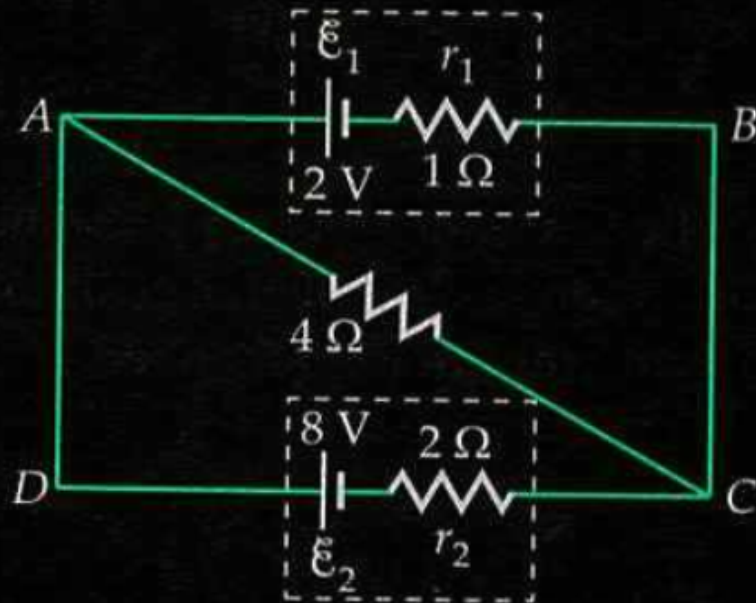
[Ans. (i) 2 A (ii) 4 V]



19. A cell of emf $\mathcal{E}_1 = 2 \text{ V}$ and internal resistance $r_1 = 1 \Omega$ is connected to another cell of emf $\mathcal{E}_2 = 8 \text{ V}$ and internal resistance $r_2 = 2 \Omega$ through an external resistance of 4Ω as shown in Fig. 3.189. Find the potential difference between point A and point C.

[CBSE 20C]

(Ans. $\frac{24}{7} \text{ V}$)



18. In Fig. 3.188, $ABCD$ is a uniform circular wire of resistance 2Ω . AOC and BOD are two wires along two perpendicular diameters of the circle, each having same resistance 1Ω . A battery of emf $\mathcal{E}$ and internal resistance r is connected between the points A and D . Calculate the equivalent resistance of the network.

(Ans. $\frac{15}{14}\Omega$)

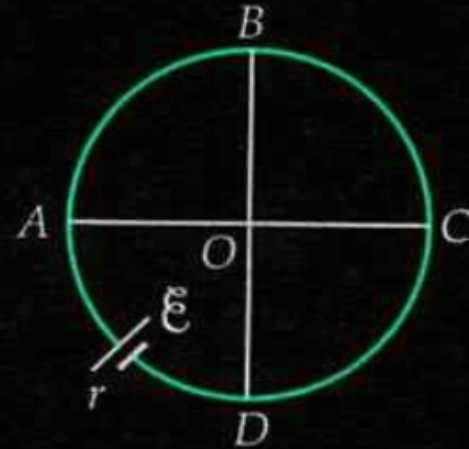


Fig. 3.188